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UltraPACCapDefTopic

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About the UltraPAC80 Capture

About the UltraPAC80 Capture
The Ultra Precision Analog Channel (UltraPAC80) Capture instrument is designed to capture waveforms for applications such as wireless baseband, audio, HDTV/video, xDSL, DVD/ODD, and set top box/cable modem device testing. Typical tests that can be performed include: gain accuracy, frequency response, distortion, signal-to-noise ratio, and linearity.
The following topics are included:

- Features
- Safety Information
- Key Terms
- UltraPAC80 Capture Software Licensing Requirements

UltraPAC80 Capture information includes the following sections:

- UltraPAC80 Capture Theory of Operation
- UltraPAC80 Capture Programming
- UltraCapture Debug Display
- UltraPAC80 Capture Examples

Related Topics

- UltraPAC80 Specifications
- UltraCapture (VBT syntax)
- UltraPAC80 DIB Design Guidelines

Eight UltraPAC80 Capture instruments reside on the UltraPAC80 instrument board with eight UltraPAC80 Source instruments, which generate AC waveforms. For more information, see About the UltraPAC80 Source.

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About the UltraPAC80 Capture : Features

Features

The UltraPAC80 Capture provides:

- 8 fully independent digitizers
  - Independent user clocks
  - Independent pattern control
  - 8 Msample depth per capture
- High Frequency (HF) Mode 75 MHz direct capture bandwidth
  - Max AC range 3Vpp single-ended, 3Vpk differential (to 30MHz), 1.5Vpk differential (to 75MHz)
- High Resolution (HR) Mode 50 kHz Direct Capture Bandwidth
  - Max AC range 5Vpp single-ended, 5Vpk differential
- Single-ended and differential measurement
- Bandwidths of 50kHz, 30MHz, and 75MHz

- Better than 56 ps channel-to-channel skew without calibration DIB
- Access to Parametric Measurement Unit (PPMU) per pin, 32 PPMUs per channel card
- Support for Instrument Initiated Move (IIM) and Background DSP
- Signals loading and triggering initiated from digital patterns
- Bidirectional operation from UltraSource out of UltraCapture for four channels per instrument board (two I/Q port pairs)
- Ability to assign each UltraPAC80 channel to a unique time domain (16 total, eight UltraSources and eight UltraCaptures)
- Ability to control each UltraPAC80 channel by unique time domains simultaneously using Sync-Link
  - Independent access to Sync-Link
  - Independent sample clock programming
  - Independent clock synchronization with other UltraPAC80 channels or with digital clocks
  - Independent control by digital pattern
  - Independent capture memory and triggers
- Pin-to-pin compatibility with BBAC and TurboAC
- BBAC-to-UltraPAC80 conversion tool to aid in test program porting
- Connections sets (CSets) that allow user defined connection setups to be applied using digital patterns

For more information, see UltraPAC80 Specifications.

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About the UltraPAC80 Capture : Safety Information

Safety Information

When operating a UltraPAC80 Capture instrument, always follow general UltraFLEX test system safety guidelines. See Test System Safety.

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About the UltraPAC80 Capture : Key Terms

Key Terms

channel	A single Instrument to DIB connection. For example, a differential pair consists of two channels. Two single-ended sites consist of two channels.
CMEM	Capture Memory
DGS	Device Ground Sense. The DGS input is tied to the DUT's analog ground plane. This ensures that all signals sourced or captured from the UltraPAC80 are referenced to device ground rather than to instrument ground (CCC).
High Frequency mode	An instrument parameter that specifies use of the High Frequency (HF) signal path, DC - 75 MHz. This path is used when 30MHz and 75MHz bandwidths are selected.
High Resolution mode	An instrument parameter that specifies use of the High Resolution (HR) signal path, DC - 50 kHz. This path is used when the 50kHz bandwidth is selected.
IIM	Instrument Initiated Move
instrument	The set of UltraPAC80 functionality that can independently source or capture a waveform. An instrument includes functionality to generate and capture a waveform, as well as the functionality to deliver or receive the waveform from multiple outputs or inputs.
ISL	Instrument Sync-Link
ksps	Kilosamples per second

Msps	Megasamples per second
offset	The voltage removed from a captured signal. This allows the signal to be centered within a capture voltage range (centered about 0V) for optimal capture performance.
PPMU	Pin Parametric Measurement Unit

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[About the UltraPAC80 Capture](#) : UltraPAC80 Capture Software Licensing Requirements

UltraPAC80 Capture Software Licensing Requirements

IG-XL enforces the following counted licenses for each UltraSource or UltraCapture channel (pos/neg channel pair):

UltraPAC80 License Enabling Points

License Name	Enabling Points
UltraSource	Waveform bandwidth in MHz: 80 (“full”), 40, 3
UltraCapture	Signal bandwidth in MHz: 75 (“full”), 40, 3

One license is required for each UltraSource or UltraCapture channel in the channel map, at a level that accommodates the highest signal frequency sourced or captured on the channel.

Online, IG-XL obtains these licenses from the counted license file. Offline, the simulated configuration file defines these counted licenses under the license enablers section.

In an offline simulated or current configuration file, these licenses appear as follows:

```
<LICENSE_ENABLERS>
#Off-line enabling levels
#FEATURE        ENABLED LEVEL    LICENSE COUNT    LEGAL ENABLING POINTS
UltraCapture    75               1                75,40,3
UltraSource     80               1                80,40,3
</LICENSE_ENABLERS>
```

IG-XL checks for UltraSource and UltraCapture licenses on validation of a test program to verify that the number of channels used in the channel map does not exceed the number licensed. If you do not have an adequate number of licenses for the number of channels, IG-XL returns an error. For example:

Unable to obtain enough UltraSource licenses to validate this job. At least 4 UltraSource licenses are required for channels in the Channel Map.

IG-XL also verifies licenses on execution of a test program to verify that the program does not source or capture signals whose frequency exceeds the licensed limitations. A run-time error occurs if the program sources or captures at frequencies higher than allowed by the licenses. For example:

Unable to obtain enough UltraCapture licenses to run this job. At least the following UltraCapture licenses are needed: 2 @ 3 MHz, 2 @ 40 MHz, 3 @ 75 MHz.

Each UltraCapture 75 MHz (“full”) license allows one channel to operate at any legal capture sample rate. Each UltraCapture 3 MHz license allows one channel to operate up to a maximum sample rate of 8 Msps. This permits the channel to capture frequencies up to a maximum of 3 MHz, due to the requirement that the capture sample rate must be 8/3 times greater than the highest frequency of interest. Similarly, each UltraCapture 40 MHz license allows one channel to operate up to a maximum sample rate of (40 * 8/3) Msps, which permits the channel to capture frequencies up to a maximum of 40 MHz. See UltraPAC80 Specifications for the minimum sample rate specifications.

Each UltraSource 80 MHz (“full”) license allows one channel to operate at any legal source sample rate. Each UltraSource 40 MHz license allows one channel to operate at any legal source sample rate up to a maximum of 160 Msps. This permits the channel to source frequencies up to a maximum of 40 MHz, due to the requirement that the source sample rate must be four times greater than the highest frequency of interest when the bandwidth setting is 80 MHz. Similarly, each UltraSource 3 MHz license allows one channel to operate at any legal source sample rate up to a maximum of 12 Msps. This permits the channel to source frequencies up to a maximum of 3 MHz. See UltraPAC80 Specifications for the minimum sample rate specifications.

For more information, see IG-XL Licensing in the IG-XL Administration and Licensing section.

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UltraPAC80 Capture Examples

UltraPAC80 Capture Examples

This section contains program examples for the UltraCapture instrument. These IG-XL workbook examples show all IG-XL data and VBT code necessary for a complete running program.

The following topics are available:

- UltraSource/UltraCapture Loopback
- UltraCapture Signals VBT Examples

For more information, see:

- UltraCapture (VBT Syntax Reference)
- UltraPAC80 Capture Programming

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UltraPAC80 Capture Examples : UltraSource/UltraCapture Loopback

UltraSource/UltraCapture Loopback

This workbook example includes the following program components:

- DataTool Sheets
- VBT Code

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UltraPAC80 Capture Examples : UltraSource/UltraCapture Loopback : DataTool Sheets

DataTool Sheets

This workbook example uses the following DataTool sheets:

- Pin Map Sheet
- Channel Map Sheet
- Test Instances Sheet
- Flow Table Sheet
- Time Sets (Basic)

Pin Map Sheet

Pin Map

Group Name	Pin Name	Type	Comment
	Src1_P	Analog	
	Src1_N	Analog	
	Cap1_P	Analog	
	Cap1_N	Analog	
	DigControl	I/O	

Channel Map Sheet

Channel Map

DIB ID: View Mode: Signal

Device Under Test		Tester Channel		
Pin Name	Package Pin	Type	Site 0	Comment
Src1_P		UltraSource	22.SrcPos1	4
Src1_N		UltraSource	22.SrcNeg1	
Cap1_P		UltraCapture	22.CapPos1	
Cap1_N		UltraCapture	22.CapNeg1	
DigControl		I/O	1.ch0	

Test Instances Sheet

Test Instances

	Test Procedure			DC Specs		AC Specs		
Test Name	Type	Name	Called As	Category	Selector	Category	Selector	Time Sets
Source_Capture_LoopBack1	VBT	Source_Capture_LoopBack						TSB

Flow Table Sheet

Flow Table

		Gate			Command			
Label	Enable	Job	Part	Env	Opcode	Parameter	TName	TNum
					Test	Source_Capture_LoopBack1		

Time Sets (Basic)

Time Sets (Basic)

Timing Mode: Single Master Timeset Name:
Time Domain:

	Cycle	Pin/Group			Data		Drive				Compare			Edge Resoluti
Time Set	Period	Name	Clock Period	Setup	Src	Fmt	On	Data	Return	Off	Mode	Open	Close	Mode
ts0	200.E-09	DigControl		i/o	PAT	NR		0			Off			Auto

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UltraPAC80 Capture Examples : UltraSource/UltraCapture Loopback : VBT Code

VBT Code

```
Public Function Source_Capture_LoopBack()
    Dim SourceSine As New DSPWave
    Dim CaptureWave As New DSPWave
    Dim i As Long
    ' Connect UltraSource to UltraCapture Loopback
    TheHdw.UltraSource.Pins("Src1_P").Connect tlUltraSourceFrom_Pos Or _
        tlUltraSourceFrom_Neg, tlUltraSourceTo_Loopback
    TheHdw.UltraCapture.Pins("Cap1_P").Connect tlUltraCaptureConnectFrom_Pos _
        Or tlUltraCaptureConnectFrom_Neg, tlUltraCaptureConnectTo_Loopback
    ' Create Sinewave Samples to Source
    Call SourceSine.CreateConstant(0, 65536)
    For i = 0 To 65536 - 1
        SourceSine.Element(i) = Sin((3 * i * 2 * Pi / 65536)) / 2
    Next i
    ' Load Setting and Samples into the UltraPAC80 and Set up Source Signal
    TheHdw.UltraSource.Pins("Src1_P").Signals.Add "SourceSignal"
    TheHdw.UltraSource.Pins("Src1_P").Signals.DefaultSignal = "SourceSignal"
    TheExec.WaveDefinitions.CreateWaveDefinition "WaveDef", SourceSine, True
    With TheHdw.UltraSource.Pins("Src1_P").Signals("SourceSignal")
        .WaveDefinitionName = "WaveDef"
        .Amplitude = 1
        .Offset = 0
        .SampleSize = 65536
        .SampleRate = 655360
```

```
        .CycleCount = 1
        .LoadSettings
    End With
    ' Set up UltraCapture
    TheHdw.UltraCapture.Pins("Cap1_P").Signals.Add "CaptureSignal"
    TheHdw.UltraCapture.Pins("Cap1_P").Signals.DefaultSignal = "CaptureSignal"
    With TheHdw.UltraCapture.Pins("Cap1_P").Signals("CaptureSignal")
        .SampleSize = 65536
        .SampleRate = 655360
        .CycleCount = 1
        .LoadSettings
    End With
    ' Start Sourcing the signal and capturing with Pattern
    'TheHdw.Digital.ApplyLevelsTiming True, True, True
    TheHdw.Patterns("./src_cap_loop.pat").Load
    TheHdw.Digital.Patgen.TimeoutEnable = False
    TheHdw.Patterns("./src_cap_loop.pat").Start "", ""
    TheHdw.Digital.Patgen.HaltWait
    ' Wait for Capture to complete
    While TheHdw.UltraCapture.Pins("Cap1_P").IsCaptureInProgress
    Wend
    ' Bind capture data to a DSPwave object
    CaptureWave = TheHdw.UltraCapture.Pins("Cap1_P") _
        .Signals("CaptureSignal").DSPWave
    CaptureWave.Plot "UltraPAC80 LOOPBACK"
    ' Do an SNR Analysis on captured data and display it in immediate window
    Debug.Print CaptureWave.Spectrum.CalcSNR
    ' Disconnect the UltraSource from the UltraCapture
    TheHdw.UltraSource.Pins("Src1_P").Disconnect tlUltraSourceFrom_Pos Or _
        tlUltraSourceFrom_Neg, tlUltraSourceTo_Loopback
    TheHdw.UltraCapture.Pins("Cap1_P").Disconnect _
        tlUltraCaptureConnectFrom_Pos Or tlUltraCaptureConnectFrom_Neg, _
        tlUltraCaptureConnectTo_Loopback
End Function
```

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UltraPAC80 Capture Examples : UltraCapture Signals VBT Examples

UltraCapture Signals VBT Examples

The following examples demonstrate UltraCapture VBT methods and properties:

- Use MST to Set Up, Load, and Trigger an UltraCapture Signal
- Report Information About a Collection of UltraCapture Signals
- Set Up and Trigger an UltraCapture Signal
- Use LoadSettings for an UltraCapture Signal and Read Programmed Settings
- Use Pattern Control with UltraCapture Default Signal

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UltraPAC80 Capture Examples : UltraCapture Signals VBT Examples : Use MST to Set Up, Load, and Trigger an UltraCapture Signal

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Use MST to Set Up, Load, and Trigger an UltraCapture Signal

This code uses a Mixed Signal Timing (MST) set to set up, load, and trigger an UltraCapture signal.

```
Public Sub UseMSTToSetupAndTriggerUltraCapSignal()
    ' Use MST to set up, load, and trigger a capture.
    ' Replace "AOUT" with your UltraCapture (pos) pin name
    Dim pinName As String
    pinName = "AOUT"
    Dim myDspWave As New DSPWave
    TheHdw.UltraCapture.pins(pinName).Signals.Add "cap1"
    With TheHdw.UltraCapture.pins(pinName).Signals("cap1")
        ' Apply a Mixed Signal Timing Set to this Test Instance before
        ' ApplyMixedSignalTiming is called.
        .ApplyMixedSignalTiming
        ' Assign the PinListData object for access to the captured data
        myDspWave = .DSPWave
        ' Use defaults for all other Signal properties
        ' Load Signal hardware values into hardware
        .LoadSettings
        ' Trigger the Signal capture
        .Trigger
    End With
End Sub
```

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UltraPAC80 Capture Examples : UltraCapture Signals VBT Examples : Report Information About a Collection of UltraCapture Signals

---

Report Information About a Collection of UltraCapture Signals

This code reports information about a collection of UltraCapture signals using Count and List.

```
Public Sub ReportInfoAboutUltraCapSignalsColl()
    ' Replace "AOUT" below with your UltraCapture (pos) pin name
    Dim pinName As String
    pinName = "AOUT"
    ' First, add 10 Signals to the Signals collection on Pin 'pinName'
    Dim i As Long
    Dim sigCount As Long
    For i = 1 To 10
        TheHdw.UltraCapture.pins(pinName).Signals.Add "signal" & CStr(i)
    Next i
    sigCount = TheHdw.UltraCapture.pins(pinName).Signals.Count
    ' Print out each Signal name
    Dim sigList() As String
    sigList = TheHdw.UltraCapture.pins(pinName).Signals.List
    Debug.Print "There are " & sigCount & " signals in the collection " & _
                "on pin '" & pinName & "'. They are:"
    For i = 0 To sigCount - 1
        Debug.Print vbCrLf & sigList(i)
    Next i
End Sub
```

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UltraPAC80 Capture Examples : UltraCapture Signals VBT Examples : Set Up and Trigger an UltraCapture Signal

Set Up and Trigger an UltraCapture Signal

This code sets up and triggers a signal.

```
Public Sub CreateAndTriggerUltraCaptureSignalWithDirectProgramming()
    ' Use direct timing programming to set up, load, and trigger a capture
    ' Replace "AOUT" with your UltraCapture (pos) pin name
    Dim pinName As String
    pinName = "AOUT"
    Dim myDSPWave As New DSPWave
    TheHdw.UltraCapture.pins(pinName).Signals.Add "cap1"
    With TheHdw.UltraCapture.pins(pinName).Signals("cap1")
        .SampleRate = 20000
        .SampleSize = 20
        ' Assign a PinListData object for access to the captured data
        myDSPWave = .DSPWave
        ' Use defaults for all other Signal properties
        ' Load hardware settings for the Signal
        .LoadSettings
        ' Trigger the capture
        .Trigger
    End With
End Sub
```

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UltraPAC80 Capture Examples : UltraCapture Signals VBT Examples : Use LoadSettings for an UltraCapture Signal and Read Programmed Settings

---

Use LoadSettings for an UltraCapture Signal and Read Programmed Settings

This code programs a new signal, uses Load Settings, and reads the programmed hardware settings.

```
Public Sub ProgramUltraCaptureSignalAndReadbackHardware()
    ' Program a new signal, Load Settings, and read the programmed hardware
    ' settings from Pins language.
    ' Replace "AOUT" with your UltraCapture (pos) pin name
    Dim pinName As String
    pinName = "AOUT"
    TheHdw.UltraCapture.pins(pinName).Signals.Add "cap1"

    With TheHdw.UltraCapture.pins(pinName).Signals("cap1")
        .SampleRate = 20000
        .SampleSize = 512
        .VoltageRange = 3.0
        .Offset.Enable = tlUltraCaptureOffsetDisable
        ' Use defaults for everything else.
        ' Load hardware settings
        .LoadSettings
    End With

    ' Now that settings are loaded, read back hardware values from Pins
    ' language
    With TheHdw.UltraCapture.pins(pinName)
```

```
        Debug.Print "Loaded SampleRate: " & CStr(.SampleRate)
        Debug.Print "Loaded SampleSize: " & CStr(.SampleSize)
        Debug.Print "Loaded Offset Enable: " & CStr(.Offset.Enable)
        Debug.Print "Loaded VoltageRange: " & CStr(.VoltageRange)
    End With
End Sub
```

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UltraPAC80 Capture Examples : UltraCapture Signals VBT Examples : Use Pattern Control with UltraCapture Default Signal

Use Pattern Control with UltraCapture Default Signal

This code uses pattern control with the DefaultSignal.

```
Public Sub UseUltraCapDefaultSignalAndPatternControl()
    ' Replace "AOUT" with your UltraCapture (pos) pin name
    Dim pinName As String
    pinName = "AOUT"
    ' Replace ".\Patterns\myPat.pat" with your pattern name
    Dim patternName As String
    patternName = ".\Patterns\myPat.pat"
    ' Add 5 Signals to the Signals collection on Pin 'pinName'.
    Dim i As Long
    Dim sigCount As Long
    Dim sigList() As String
    Dim myDSPWave As New DSPWave
    For i = 1 To 5
        TheHdw.UltraCapture.pins(pinName).Signals.Add "signal" & CStr(i)
        With TheHdw.UltraCapture.pins(pinName).Signals("signal" & CStr(i))
            ' Apply a Mixed Signal Timing Set to this Test Instance before
            ' ApplyMixedSignalTiming is called.
            .ApplyMixedSignalTiming
            ' Assign a PinListData object for access to the captured data
            myDSPWave = .DSPWave
        End With
    Next i
```

```
    ' Set each Signal to be default signal and burst pattern.
    For i = 1 To 5
        With TheHdw.UltraCapture.pins(pinName).Signals
            ' Load Settings
            .Item("signal" & CStr(i)).LoadSettings
            ' Set the DefaultSignal
            .DefaultSignal = "signal" & CStr(i)
        End With
        ' This pattern should have an unnamed "Trig" microcode on pin
        ' 'pinName'.
        ' Run "" will start pattern from the beginning and wait for
        ' completion.
        TheHdw.Digital.Patterns.Pat(patternName).Run ""
    Next i
End Sub
```

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UltraCapture Debug Display

UltraCapture Debug Display

The UltraCapture Debug Display is an interactive diagram of the UltraPAC80 Capture hardware and signal path which shows the instrument's settings and connections in the context of a site. You can change the values, options, and settings in the display and run your test program to see the effects of the changes.

The following topics are available:

- Starting the UltraCapture Debug Display
- UltraCapture Debug Display Interface
- Using the UltraCapture Debug Display

All UltraPAC80 instruments are also displayed on the UltraPac Connections display. For more information, see Using the UltraPac Connections Debug Display.

This section describes features specific to the UltraCapture Debug Display. For more information on using debug displays, see the following:

- Using Debug Displays in TDE
- Producing Code from Debug Displays

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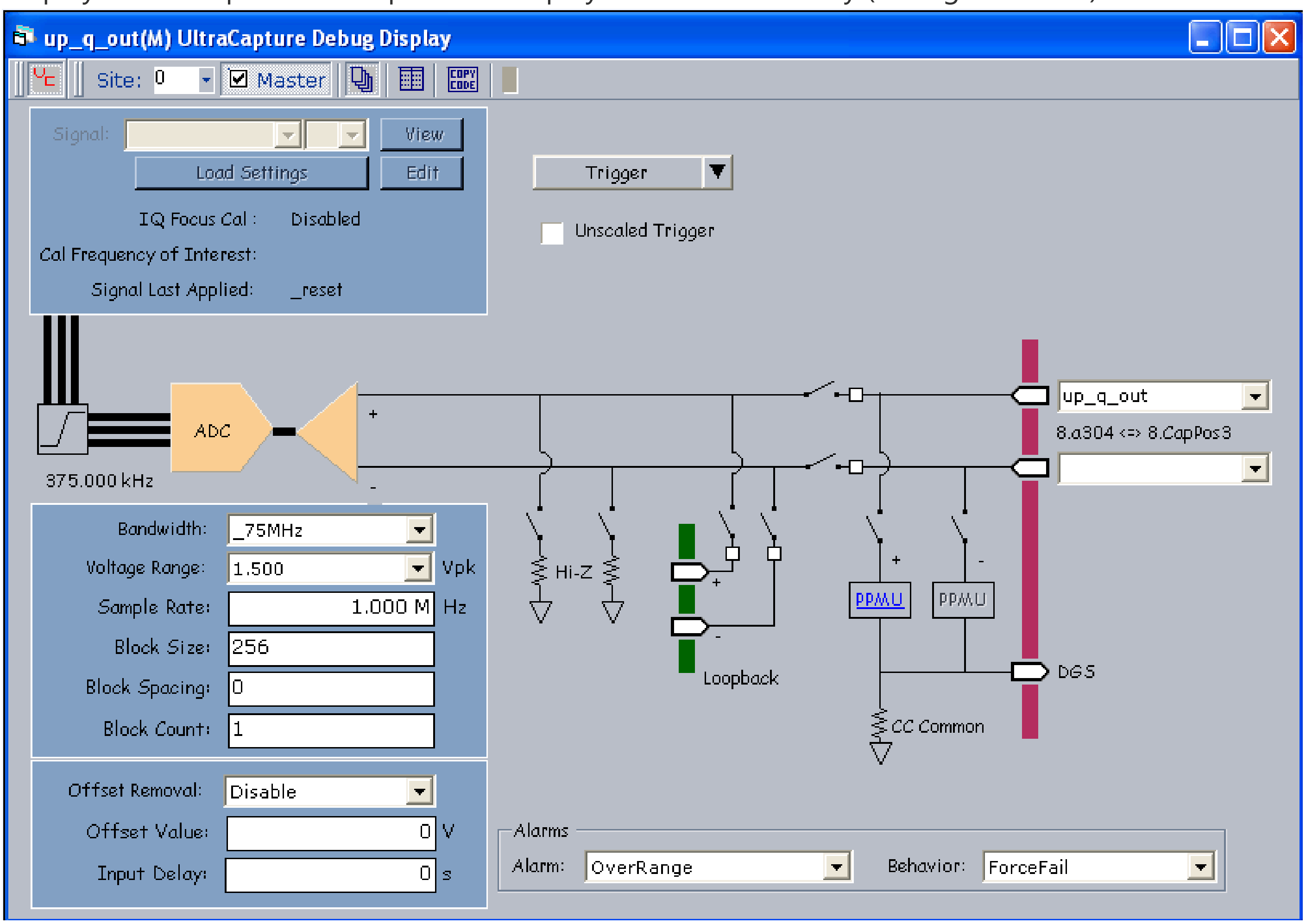
UltraCapture Debug Display : Starting the UltraCapture Debug Display

Starting the UltraCapture Debug Display

To start the UltraCapture debug display for an UltraPAC80 Capture instrument, first validate an UltraPAC80 test program. Then, on the IG-XL Tools tab, in the Launch group, select Debug Displays>UltraCapture. This opens the display for active sites only (see figure below).



UltraCapture Debug Display

If you start a debug display before the job has been validated, an error appears in the display saying that the program has not been validated. If you open a display for an instrument that is in the configuration file but not in the channel map, the display shows a message that no instruments have been found.

Once the job is validated, the display shows the current hardware settings. When you run to a breakpoint in the flow or in the VBT code, or as you step through the flow or code, IG-XL updates the display to show any changes to the settings.

For information on other methods for starting debug displays, see Starting Debug Displays in the Test Debug Environment (TDE) section.

For information on fields used in the UltraCapture debug display, see UltraCapture Debug Display Interface.

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UltraCapture Debug Display : UltraCapture Debug Display Interface

UltraCapture Debug Display Interface

A trigger on the debug display functions as a trigger on an oscilloscope, in that it makes a capture using the current instrument debug display settings. Captured data from triggering the debug display are for visual reference only. You cannot use the data from the debug display capture for processing because the captured samples are not moved through IIM to the Support Board. Also, the debug display does not store the data. Once you have displayed it, therefore, you cannot retrieve it.

The table lists UltraCapture Debug Display menus, buttons, and fields.

UltraCapture Debug Display Fields

Menu, Button, or Field	Action
Signal	All user-defined signal names appear in the Signal field. When a signal is selected, the Load Settings button is enabled. When a capture is moved to the Support Board and it has not been redirected to a DSP Procedure, that signal's name is marked with an asterisk (*). You can view such a signal in WaveScope. If multiple captures on the Support Board are associated with the same Signal name, the box between the Signal field and the View button is enabled, and you can select which capture you are interested in viewing.
View	This button is enabled when you select a signal containing an asterisk. Viewing a signal only makes a copy of the samples on the Support Board. The data is not removed from the Support Board.

Edit	Brings up the PSet and Signals Editor (PSSE) with the selected signal loaded. This control is grayed out if no signals are available. See PSet and Signals Editor in the TDE section.
Load Settings	This button is enabled when a signal name is selected from the Signal field. It loads the settings of the specified Signal. That signal must be defined from VBT.
IQ Focus Cal	This read-only field indicates whether IQ calibration is Enabled or Disabled.
Cal Frequency of Interest	This read-only field indicates the frequency of interest for calibration of a signal.
Signal Last Applied	This read-only field indicates the name of the last signal applied.
Trigger	This menu selects a type of trigger: • Trigger: This selection functions similarly to a single trigger button on an oscilloscope. • Multi-Pin Trigger: This selection allows you to trigger more than one pin. If you select Multi-Pin Trigger with Open in Single Wavescope checked, then captured waves are displayed in a single WaveScope. Otherwise, they are displayed in multiple WaveScopes. You can set the sample rate, block count, block size, block spacing, and extra samples on the display and then click Trigger to make a capture. The trigger is disassociated from the Signal name. No signal must exist for the Trigger button to function. Captures made from the display are shown on WaveScope, and the data is then discarded. The display does not move data to the Support Boards.
Unscaled Trigger	This field allows you to enable the Trigger button to display an unscaled capture waveform in the WaveScope. The samples will vary from 1, Indicating the full scale range of the capture ADCs.
Bandwidth	The bandwidth of the capture: 50kHz, 30MHz, or 75MHz.
Voltage Range	The voltage range of the capture. The selection of voltage ranges depends on the Bandwidth.

	• Minimum: 0.0V • Maximum: 5V (50kHz) or 3V (30MHz or 75MHz)
Sample Rate	The number of samples to capture per second. • Minimum: 5.000ksps • Maximum: 200Msps
Block Size	The number of samples per block. • Minimum: 1 • Maximum: 1 048 565 See Capture Sample Size Parameters.
Block Spacing	The number of samples to skip per block. • Minimum: 0 • Maximum: 4 294 967 295 See Capture Sample Size Parameters.
Block Count	The number of blocks to capture. • Minimum: 1 • Maximum: 4095 See Capture Sample Size Parameters.
Offset Removal	Assigns an offset to the Capture pins. The options are: • Disable

	• PosEnable • NegEnable • PosAndNegEnable Offset circuitry is added to the display according to the assignment selected.
Offset Value	Sets the amount offset to apply. • Minimum: -6.0V • Maximum: 6.0V
Input Delay	Sets the delay in seconds associated with an UltraCapture input path.
PPMU	Click this button to open the PPMU debug display for the selected pin. See PPMU Debug Display.
Alarm	Selects the alarm for which to set the Behavior. The options are: • All • Over Range Selecting an alarm only affects the alarm for that pin. Note that All and Over Range perform the same function. Also, the special option All is write-only, so the behavior displayed for it is not updated if it is changed using VBT.
Behavior	Selects the action that should occur if the selected alarm condition exists for the pin. The options are: • Default • Continue

- Off
- Force Bin
- Force Fail

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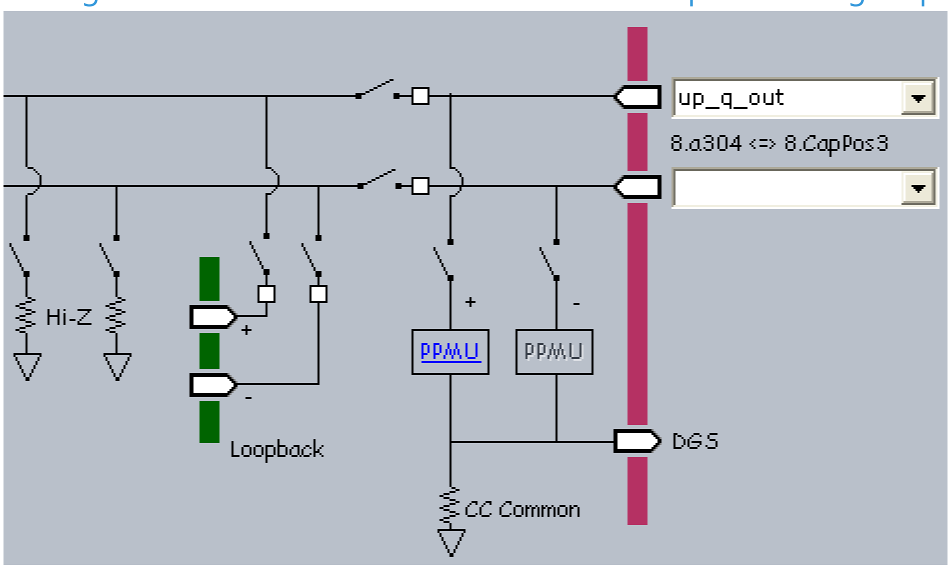
UltraCapture Debug Display : Using the UltraCapture Debug Display

Using the UltraCapture Debug Display

The figure shows the connections on the UltraCapture Debug Display.



UltraCapture Debug Display Connections

The following topic describes the interface behavior and the actions that you perform to make channel and resource connections in the UltraCapture debug display:

- Opening and Closing a Relay Path

The following sections describe the resources that the UltraCapture channels can be connected to:

- Connecting DUT Pins
- Connecting the Internal Analog Loopback Path

- Connecting a PPMU

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UltraCapture Debug Display : Using the UltraCapture Debug Display : Opening and Closing a Relay Path

Opening and Closing a Relay Path

The UltraCapture debug display simulates channel connections to the DUT and other resources. Graphically these connections are represented by open and closed relays along the connection path. However, clicking a relay does not open or close it.

To open or close a relay path:

1. Right-click the node along the signal path. A menu appears and lists the connect or disconnect option for that path.
2. Select the connection destination. The relay opens or closes to make or break the connection.

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UltraCapture Debug Display : Using the UltraCapture Debug Display : Connecting DUT Pins

Connecting DUT Pins

The UltraCapture debug display allows you to connect an instrument channel to the DUT pins specified in the test program's channel map.

Starting the display automatically reads in the pins and lists them in the DUT pin drop-down lists.

The DIB channel string is also read into the display and is shown beside the appropriate instrument channel.

To connect an instrument channel or resource to a DUT pin:

1. Right-click the DUT node.
2. Select the channel from the menu.

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UltraCapture Debug Display : Using the UltraCapture Debug Display : Connecting the Internal Analog Loopback Path

Connecting the Internal Analog Loopback Path

The loopback connection is provided for debugging. It allows you to connect an UltraSource directly to an UltraCapture, allowing them to measure the sourced waveform.

The UltraPAC80 instrument board has eight instrument pairs (for example, UltraCapture and UltraSource in Channel 1). An UltraCapture Positive loops back to an UltraSource Positive Loopback is paired src/cap 1/2, 3/4, 5/6, and 7/8. Pos can only connect to Pos and Neg to Neg.

You can control the Pos and Neg channels independently. For example, you can loop back Positive or Negative or Both.

To connect an UltraSource to an UltraCapture for loopback purposes:

1. Right-click the Loopback Positive node and select Connect to Capture Pos.
2. Right-click the Loopback Negative node and select Connect to Capture Neg.
3. Open the UltraSource debug display.
4. Right-click the Loopback Positive node and select Connect to Source Pos.
5. Right-click the Loopback Negative node and select Connect to Source Neg.

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UltraCapture Debug Display : Using the UltraCapture Debug Display : Connecting a PPMU

Connecting a PPMU

You can connect each UltraCapture channel to a Pin Parametric Measurement Unit (PPMU) circuit for performing continuity tests.

Although the UltraCapture display shows the channel connections to the PPMU circuits, you make PPMU connections in the PPMU debug display, not in the UltraCapture display.

To connect an UltraCapture channel (Pos, Neg) to a PPMU circuit:

1. Click the appropriate PPMU hyperlink 

on the UltraCapture Debug Display based on the UltraCapture channel that is to be connected. The PPMU debug display starts.

2. Select the pin to be connected to the PPMU from the Pin menu.

3. Click the relay to close the connection.

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UltraPAC80 Capture Programming

UltraPAC80 Capture Programming

This section describes programming of an UltraCapture instrument to capture data from a DUT.

The following topics are available:

- Simulated Configuration Files
- Programming Concepts
- Programming Overview
- UltraPAC80 DataTool Programming
- UltraCapture Pattern Control
- Using UltraPAC80 in Concurrent Testing
- Achieving IQ Alignment for UltraCapture
- Programming the Relay Counter

IG-XL also includes tool for converting BBAC test programs to UltraPAC80. For more information, see Using the BBAC to UltraPAC Converter Tool in the UltraPAC80 Source Programming section.

For more information, see:

- UltraCapture (VBT Syntax Reference)
- UltraPAC80 Capture Examples

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UltraPAC80 Capture Programming : Simulated Configuration Files

Simulated Configuration Files

You can use an UltraCapture instrument offline by forcing IG-XL to map the UltraPAC80. Do this by creating a SimulatedConfig.txt file and placing the file in any of the following locations:

- Folder containing the IG-XL workbook
- \tester folder: C:\Program Files (x86)Teradyne\IG-XL\<version>\tester
- \bin folder: C:\Program Files (x86)\Teradyne\IG-XL\<version>\bin

The following SimulatedConfig.txt file maps an UltraPAC80 board in slot 15:

```
<Version>
1
</Version>

<TEST_HEAD TH 1>
#slot[.subslot]   Type            idprom (type serial rev company)

15.0        UltraPAC        605-743-00 0000000 0000-A 5445
            UltraPAC        939-420-01 0000000 0000-A 5445
            UltraPAC-0      939-392-00 0000000 0000-A 5445
            UltraPAC-1      939-392-00 0000000 0000-A 5445
            PowerBoard      604-207-00 0000000 0000-A 5445

24.0        SupportBoard    974-220-12 0000000 0000-A 0000
```

SupportBoard 956-354-00 0000000 0000-A 0000

62.0 DistributionBoard 956-355-00 0000000 0000-A 0000
DistributionBoard 956-355-00 0000000 0000-A 0000

</TEST_HEAD TH 1>

To use an UltraCapture offline, the SimulatedConfig.txt file must include every board needed in the IG-XL workbook. The SimulatedConfig.txt file represents the state of the offline tester. To use the UltraCapture online, you need not edit any files. IG-XL automatically maps all boards in the tester. For more information, see Tester Configuration.

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UltraPAC80 Capture Programming : Programming Concepts

Programming Concepts

The following topics are available:

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Differential and Single-Ended Measurements

To make differential DUT measurements with an UltraCapture, connect both the positive and negative pins of a common capture instrument to the device differential output. Select the appropriate input voltage range. The voltage range is in units peak differential voltage when measuring differential signals.

To make single-ended DUT measurements with an UltraCapture, connect only one of the positive or negative pins of the capture instrument to the device single ended output. Leave the corresponding pin (positive or negative) for that instrument channel disconnected from the DUT. In this way, the UltraPAC hardware will automatically reference the single-ended measurement relative to DGS. Select the appropriate input voltage range. The voltage range is in units peak single-ended voltage when measuring single-ended signals.

Note that voltage ranges for differential and single-ended measurements are different depending on the Bandwidth setting (high frequency or high resolution modes). Do not use the highest two voltage ranges (3.536 Vpk and 5.0 Vpk for the HR path and 2.121 Vpk and 3.0 Vpk for the HF path) when making single-ended measurements as compliance limits prohibit fully using these ranges without clipping the analog front end amplifiers.

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UltraPAC80 Capture Programming : Programming Concepts : Capture Sample Size Parameters

Capture Sample Size Parameters

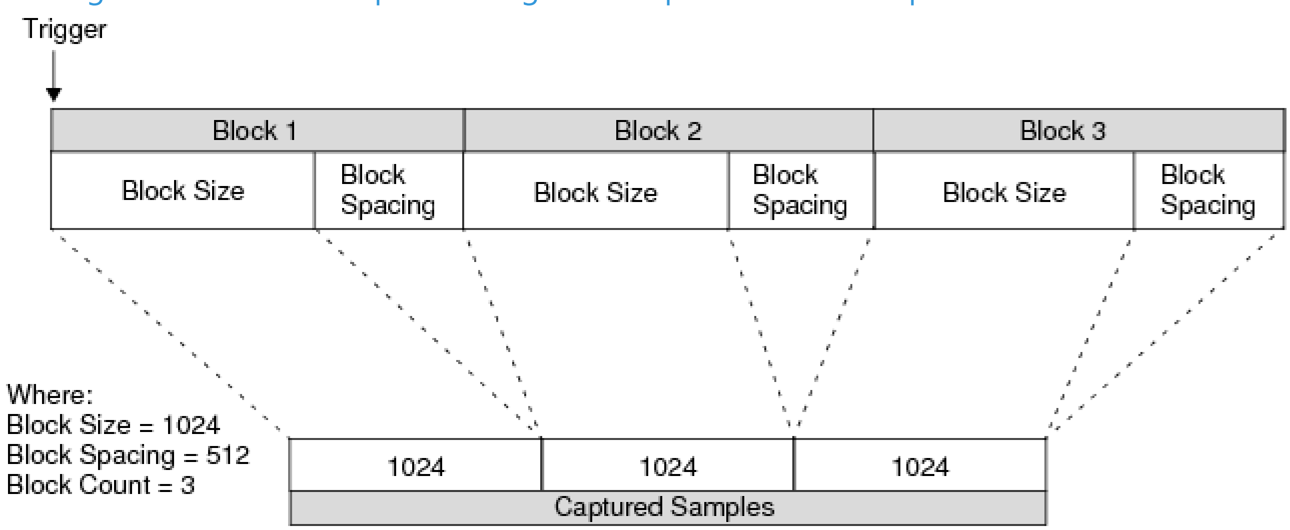
You can program UltraCapture sample size for a single capture or for a block capture. When performing a single capture, use the SampleSize parameter to specify the number of samples to capture.

When performing a block capture, the BlockSize and BlockSpacing parameters work with the BlockCount parameter to capture specified sets of samples. The BlockSize parameter specifies the number of samples in each of the one or more blocks you want to capture. The BlockSpacing parameter specifies how many samples are to be skipped (not captured in memory) between each pair of blocks captured. The BlockCount parameter specifies the number of blocks to capture.

The total capture size is calculated as:

Total Capture Size = BlockSize * BlockCount samples

This figure shows an example of using block capture mode to capture three blocks of data.



Capture Sample Size Parameters

☑ **Note:**

The BlockSize parameter is an alias of the SampleSize parameter. The SampleSize syntax is more commonly used when block capture mode is not required.

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UltraPAC80 Capture Programming : Programming Concepts : Condition Bit Programming

Condition Bit Programming

An UltraCapture instrument can produce a condition bit when a capture is complete. This allows a pattern to enter a capture loop so that it can keep running (producing DUT clocks, digital I/O, and so on) until the capture completes. When the capture completes, the pattern can either halt or set a flag to indicate to the host computer that the capture is finished. You can then suspend execution of the test program until the capture completes, at which time it is safe to continue on to set up the next test.

This use of the condition bit allows for flexible capture sizes and is easier than the error-prone technique of counting pattern vectors and converting into an equivalent time duration to determine which vector will be executing when the digitizer completes its capture. Another common technique is to use a fixed loop size in the pattern, but this does not always work with different capture timing solutions. A fixed loop size can exit too early if the capture is longer than expected, or it can be inefficient and run too long if the capture is shorter than expected.

Each UltraCapture instrument can have its own microcode column in a pattern. Using the Enable_Inst_Cond and Disable_Inst_Cond microcodes, you can open and close a window that allows the end of capture instrument condition pulse to move to the pattern generator. This allows you to dynamically select which instrument delivers a condition bit during a pattern burst. Note that if the window is not open at the time when the capture completes, the pulse is missed.

All UltraFLEX test system instrument condition bits are OR'ed together before being sent to the pattern generator. Therefore, only one instrument should have its condition bit window enabled at a time to ensure that the specific instrument has completed its capture, or reached another state that sets its condition bit.

Related Topics

- UltraCapture Pattern Control

- About PatternTool

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UltraPAC80 Capture Programming : Programming Concepts : Internal Analog Loopback

Internal Analog Loopback

The UltraPAC80 includes four loopback paths, allowing you to source and capture signals between instruments on the board. The following groups of instruments share a single loopback path:

- UltraSource 1 and 2 and UltraCapture 1 and 2
- UltraSource 3 and 4 and UltraCapture 3 and 4
- UltraSource 5 and 6 and UltraCapture 5 and 6
- UltraSource 7 and 8 and UltraCapture 7 and 8

You can use the UltraPAC Connections debug display to show loopback connections. This loopback feature can be used for verifying instrument performance and setup conditions without requiring a DIB or external instrumentation.

The following code demonstrates connecting an UltraSource to UltraCapture in loopback:

```
Public Function CloseUltraLoopbackRelay(SrcPin As String, _
                        CapPin As String) As Long
  TheHdw.UltraCapture.Pins(CapPin).Connect tlUltraCaptureFrom_Inst, _
                            tlUltraCaptureTo_LoopBack
  TheHdw.UltraSource.Pins(SrcPin).Connect tlUltraSourceFrom_Inst, _
                            tlUltraSourceTo_LoopBack
End Function
```

The following code demonstrates disconnecting an UltraSource from UltraCapture in loopback:

```
Public Function OpenUltraLoopbackRelay(SrcPin As String, _
                                    CapPin As String) As Long
    TheHdw.UltraCapture(CapPin).Disconnect tlUltraCaptureFrom_Inst, _
                                    tlUltraCaptureTo_LoopBack
    TheHdw.UltraSource(SrcPin).Disconnect tlUltraSourceFrom_Inst, _
                                    tlUltraSourceTo_LoopBack
End Function
```

☑ **Note:**

Make sure that the UltraSource and UltraCapture are disconnected from other signal paths (for example, from the DIB/DUT) to avoid external signal interference while using internal loopback. For more information, see Using the UltraPac Connections Debug Display.

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UltraPAC80 Capture Programming : Programming Concepts : Setting Bandwidth

Setting Bandwidth

You can set UltraCapture instruments to the following bandwidths:

tlUltraCaptureBandwidth__50kHz	50 kHz bandwidth through the high resolution signal path. The 50kHz antialias low-pass filter is enabled.
tlUltraCaptureBandwidth__30MHz	30 MHz bandwidth through the high frequency signal path. The 30MHz antialias low-pass filter is enabled.
tlUltraCaptureBandwidth__75MHz	75 MHz bandwidth through the high frequency signal path. The 75MHz antialias low-pass filter is enabled. Default.

The bandwidths have different ranges for SampleRate, Offset, and VoltageRange. Compared to the high frequency signal path, the high resolution signal path has:

- A lower maximum sample rate
- A larger range of offset removal
- Higher voltage ranges

Because of the exclusive nature of SampleRate, CommonMode, and VoltageRange with respect to bandwidth, a change in bandwidth may change the state of other parameters. When switching between the bandwidths, IG-XL does not return an error if the currently programmed sample rate, common mode, or voltage range is beyond the range of the newly programmed bandwidth. Instead, IG-XL reprograms the properties to the smallest or largest values valid in that bandwidth. The Bandwidth parameter is unique in this respect. Other VBT parameters return errors if a programmed state is invalid because of another parameter's state.

To eliminate invalid transitional instrument states and improve throughput, use the signals method of programming rather than specifying individual settings using .Pins programming.

For more information, see:

- UltraPAC80 Specifications
- Bandwidth (VBT Syntax Reference)

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UltraPAC80 Capture Programming : Programming Concepts : Synchronous Triggering of Multiple Instruments

Synchronous Triggering of Multiple Instruments

To synchronize triggering of multiple instruments you can use the synchronizeChannels option of the .Trigger parameter. If you specify synchronizeChannels=True, IG-XL triggers all selected instruments synchronously. This is analogous to programming the Resync and Trigger microcodes using pattern control, except that this method does not require pattern control to achieve synchronization between UltraPAC source channels. However, subsequent changes to the sample rate (using .SampleRate or .LoadSettings) will disrupt this synchronization.

Note that to use this method you must specify multiple pin names in a single .Signals.Trigger command. Therefore, you must use the same signal name for every pin that is to be started simultaneously. Individual signal properties may be different on each pin, however, but the signal name must be the same.

For more information, see Trigger in the VBT Syntax Reference.

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UltraPAC80 Capture Programming : Programming Concepts : UltraCapture Reset Values

UltraCapture Reset Values

The table lists the local reset values for the UltraCapture. Pre-job reset, post-job reset, power-up reset, and user reset all result in the same state.

Reset Values for UltraCapture Settings

Setting	Reset Value
Alarms	tlAlarmDefault
Block Count	1
Block Size/Sample Size	256
Block Spacing	0
Condition Bit Enabled	True
Input Delay	0.0s
Offset	Disabled and 0.0V
Pattern Microcode Control	tlOff
Sample Rate	1.0MHz
Bandwidth	75 MHz
Termination	HiZ
Voltage Range	1.5Vpk

Reset Values for UltraCapture Signal Settings

Setting	Reset Value
Cycle Count	1
Flatness Correction	None
IQ Calibration	Disabled
Calibration Frequency of Interest	1000 Hz

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UltraPAC80 Capture Programming : Programming Concepts : Setting Distortion (Linearity) Calibration Range

Setting Distortion (Linearity) Calibration Range

UltraPAC80 instruments (UltraSource and UltraCapture) use a linearity calibration lookup table to improve harmonic distortion over a wide range of fundamental frequencies and amplitudes. A number of tables are available for different selections of bandwidth, voltage range, connection types, and frequency of interest to be sourced.

Since IG-XL does not know the frequency of interest, you must supply this information so that the correct calibration table is applied. To specify the frequency of interest, set the property Signals.Item.Calibration.FrequencyOfInterest to the Fi that it to be sourced given that signal setup. For multitone or complex waveforms, set .FrequencyOfInterest to the highest frequency (highest tone) that is to be sourced.

For more information, see FrequencyOfInterest in the VBT Syntax Reference.

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UltraPAC80 Capture Programming : Programming Concepts : Connection Sets

Connection Sets

A connection set (CSet) is a named setup for UltraPAC80 instruments. CSets are defined in VBT and then executed from a pattern or from VBT to connect and disconnect UltraSources and UltraCaptures.

CSets provide the following advantages:

- Fast connection times
- Transparent test program modularity
- No need to disconnect at the end of every test
- Initiate every instrument setup entirely from a pattern burst

CSets are a more efficient means to set an instrument's connections. They provide higher throughput and require less relay switching, while still maintaining test independence.

A program that uses CSets for connection programming need not disconnect the instrument from the DUT at the end of each test to maintain test independence. Each test can use a connection set to ensure the connections it needs are made without regard to the previous state of the instrument.

Creating a New Connection Set

To add a CSet, use the ConnectionSets.Add syntax. For example, to add a new CSet called MyConnect, you would use:

```
TheHdw.UltraCapture.Pins().ConnectionSets.Add("MyConnect")
```

Defining a Connection Set

You define a CSet by adding connections to an existing Connection Set. When you add connections, you can define the origin of each connection and its destination. Origins can include:

- Positive and negative pins
- PPMU positive and negative pins

The destination of all connections in a CSet is the DUT.

For example, for the UltraCapture assigned to SecondPin, add a new connection to the CSet MyConnect from the positive pin to the DUT:

```
TheHdw.UltraCapture.Pins("SecondPin").ConnectionSets.Item("MyConnect"). _
   AddConnection(tlUltraCaptureCSetFrom_Pos, tlUltraCaptureCSetTo_DUT)
```

You can call this method multiple times on one CSet to build up a list of connections.

CSets are an alternative to the Pins().Connect syntax. When applying a CSet, current connections are not considered. The UltraCapture instrument handles any disconnects and other instrument operations to ensure that the hardware ends up in the specified state.

IG-XL provides the following predefined CSets for UltraCapture:

ToDUT	On the selected instruments, disconnects any existing connections and connects the selected instruments to the DUT.
Disconnect	On the selected instruments, disconnects any existing connections. Used by the Disconnect pattern microcode

You cannot modify these CSets.

Applying a Connection Set

You apply CSets using either VBT syntax or using microcode from a pattern.

From a pattern, use the Connect microcode. For example, to apply the predefined CSet ToDUT, the command would be:

```
Connect ToDUT
```

On the selected instruments, this command disconnects any existing connections and connects the instruments to the DUT.

You can also apply CSets using VBT language. For example:

```
' Apply a CSET to make the connection
TheHdw.UltraCapture.Pins("SrcPosI, SrcPosQ") _
   .ConnectionSets("CSET_DIFF_TO_DUT").Apply
```

IG-XL provides additional syntax that allows you to control and monitor CSets.

Connections Set VBT Example

The following is a CSets VBT programming example:

```
' Create Differential CSET
With TheHdw.UltraCapture.Pins("SrcPosI, SrcPosQ").ConnectionSets _
    .Add("CSET_DIFF_TO_DUT")
    .AddConnection tlUltraCaptureCSetFrom_Pos, tlUltraCaptureCSetTo_DUT
    .AddConnection tlUltraCaptureCSetFrom_Neg, tlUltraCaptureCSetTo_DUT
End With
' Create Single-Ended POS CSET + PPMU on neg
With TheHdw.UltraCapture.Pins("SrcPosI, SrcPosQ").ConnectionSets.Add("CSET_SE_AND_PPMU") _
    .AddConnection tlUltraCaptureCSetFrom_Pos, tlUltraCaptureCSetTo_DUT
    .AddConnection tlUltraCaptureCSetFrom_PPMUNeg, tlUltraCaptureCSetTo_DUT
End With
' Apply a CSET to make the connection
TheHdw.UltraCapture.Pins("SrcPosI, SrcPosQ"). _
    ConnectionSets("CSET_DIFF_TO_DUT").Apply
```

For more information, see Pins.ConnectionSets in the VBT Syntax Reference.

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UltraPAC80 Capture Programming : Programming Overview

Programming Overview

To develop a test program that uses an UltraCapture instrument, follow this basic procedure:

1. Create a pin map with pins of the DUT (Programming the Pin Map)
2. Create a channel map that allocates the tester resources to the DUT pins (Programming the Channel Map).
3. Create a mixed signal timing set (Creating a Timing Set Using the Mixed Signal Workshop).
4. Create a test flow, the order in which the test will execute, in the Flow Table sheet.
   - Create a test function using VBT code in the Visual Basic editor.
   - Instantiate the VBT test and set the test values in the Instance Editor.
5. Validate and run the test program.

For more information, see UltraCapture in the VBT Syntax Reference.

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UltraPAC80 Capture Programming : UltraPAC80 DataTool Programming

UltraPAC80 DataTool Programming

This section describes how to program several DataTool worksheets to handle basic UltraCapture parameters. The following topics are included:

- Programming the Pin Map
- Programming the Channel Map
- Creating a Timing Set Using the Mixed Signal Workshop

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UltraPAC80 Capture Programming : UltraPAC80 DataTool Programming : Programming the Pin Map

Programming the Pin Map

A pin map describes the device pins by assigning a pin type to each device pin name.

For the UltraCapture, the connections that you enter in the pin map are the analog channel connections to the device under test. Select the Analog pin type for each device pin name connected to an UltraCapture instrument. Analog is the only valid pin type for UltraCapture pins.

For more information, see Pin Map Sheet in the DataTool section.

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UltraPAC80 Capture Programming : UltraPAC80 DataTool Programming : Programming the Channel Map

Programming the Channel Map

A Channel Map sheet assigns individual device pin names to test system channels and power supplies. For an UltraCapture instrument, select the UltraCapture channel type from the Type column menu for each pin or pin group.

You must enter the tester channel information value in the Site column for each site to connect a DUT pin to the instrument. The table lists the available UltraCapture channels. Note that in the table, x indicates the DIB slot number of the UltraCapture.

UltraCapture Channel Resources

UltraPAC80 Capture Pogo Pin	DIB Channel	Signal Name
UltraCapture Pos 1	x.b405	x.cappos1
UltraCapture Neg 1	x.b404	x.capneg1
UltraCapture Pos 2	x.f405	x.cappos2
UltraCapture Neg 2	x.f404	x.capneg2
UltraCapture Pos 3	x.b405	x.cappos3
UltraCapture Neg 3	x.b404	x.capneg3
UltraCapture Pos 4	x.f405	x.cappos4
UltraCapture Neg 4	x.f404	x.capneg4
UltraCapture Pos 5	x.b405	x.cappos5
UltraCapture Neg 5	x.b404	x.capneg5

UltraCapture Pos 6	x.f405	x.cappos6
UltraCapture Neg 6	x.f404	x.capneg6
UltraCapture Pos 7	x.b405	x.cappos7
UltraCapture Neg 7	x.b404	x.capneg7
UltraCapture Pos 8	x.f405	x.cappos8
UltraCapture Neg 8	x.f404	x.capneg8

DGS pins need not be called out in a pin map or a channel map. DGS pins are automatically handled by the UltraPAC instrument when other connections are made.

For more information, see:

- Channel Map Sheet
- Connectivity Options

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UltraPAC80 Capture Programming : UltraPAC80 DataTool Programming : Creating a Timing Set Using the Mixed Signal Workshop

Creating a Timing Set Using the Mixed Signal Workshop

The major function of an UltraPAC80 Capture instrument is to capture analog waveforms from the DUT. The UltraPAC80 Capture instrument uses timing sets to specify capture sampling parameters such as sample rate and sample size.

The Mixed Signal Workshop (MSW) tool helps you to build reusable mixed signal timing solutions that meet coherence requirements. The output of the MSW is a mixed signal timing set which contains key capture sampling parameters such as sample rate and sample size. Timing sets are stored on a Mixed Signal Timing sheet in the IG-XL workbook and become a reusable timing solution for future tests.

To create a timing set:

1. Insert and set up a Mixed Signal Timing sheet in the IG-XL workbook.

    a. Select Insert>IG-XL Worksheet in the Sheet group on the IG-XL Main tab.

    b. Select Mixed Signal Timing from the Sheet type menu in the Insert IG-XL Worksheet dialog box.

    c. Type a name for the sheet in the Sheet name box (optional). Default name is Mixed Signal Timing.

    d. Click OK.

e. Type a timing set name in the Set Name column and press Enter.

f. Right-click the timing set name and select Edit from the context menu to start the Mixed Signal Workshop (MSW) tool in TDE.

The MSW display appears and displays the DUT and a capture resource block.

2. If an UltraPAC80 Capture was not automatically added to the MSW display, add an UltraPAC80 Capture resource to the MSW display. Otherwise, proceed to step 3.

a. Select MSW>Add Capture.

b. Select UltraCapture from the Select Resource dialog box.
Leave the Resource ID box empty, unless this timing set is using additional UltraPAC80 Capture instruments. If that is the case, assign the current capture instrument a unique ID name to distinguish it from the others. Note that the Resource ID must begin with a letter and can only contain letters, numbers, or the underscore character.

c. Click OK in the Select Resource dialog box.
The UltraPAC80 Capture resource block is added to the MSW display.

☑ **Note:**
MSW launches a blank set with a default Capture instrument resource. If you do not require this resource, select the block, then right-click. Choose Delete instrument from the menu.

3. Specify Capture timing. Typically, the objective of programming an UltraPAC80 Capture instrument is to find a sample rate (Fs) and sample size (N) that provide the required bandwidth, frequency resolution, and (if necessary) signal coherence.

a. Click the Solve button in the Capture resource block to start the Solver tool.

b. Select the Fs radio button to solve for the sample rate (Fs).

c. In the Other Solution Criteria frame, type in the value for the minimum capture bandwidth of interest. Typically, for sine wave testing you also select the M,N Relatively Prime check box.

d. In the Number of Cycles frame, specify a range for the bin of the fundamental frequency. Typically for sine wave testing, choose the number of cycles (M) to be odd to facilitate finding a timing solution that has M and N mutually prime.

e. In the Sample Count frame, specify a range for the capture's sample size. Typically for sine wave testing, choose the number of samples (N) to be a power of two to facilitate finding a timing solution that has M and N mutually prime.

f. In the Repeat Frequency frame, specify the expected capture frequency (FR).

☑ **Note:**
In many cases, the capture timing solution (Fs and N) actually constrains the value for a coherent source frequency (FR). The description in step 3 assumes that this is not the case and that the capture instrument is being programmed to coherently capture a known source frequency.

g. Click the Show Choices button, select a solution from the table, and click OK.
The best solution can depend on the specifics of a test's needs. Note that if no solutions are found, you must modify the inputs to the Solver. Typically, this is the sample size range or cycle count range that was used. Additionally, you can select the Show All check box to see why some of the solutions were not valid.

4. Save the timing set by clicking the Save to Excel button in the toolbar.

☑ **Note:**
If you know the desired value for the capture sample rate (Fs) and sample size (N), you can program them directly into the capture resource block fields without using the Solver. When not using the Solver, be sure to specify a capture sample rate that provides sufficient bandwidth.
For more information, see Mixed Signal Workshop.
Loading Signals
You can load signals for an UltraCapture using:

- VBT .LoadSettings syntax

- LoadSettings Microcode in a pattern

The process is as follows:

1. Define a signal and its parameters for an UltraCapture in a VBT statement. This creates a signal data structure on the host PC.

For example, the following statement defines the signal MySignal for the pin MyPin and includes sample rate and sample size:
With Thehdw.UltraCapture.Pins(“MyPin”).Signals(“MySig”)
.SampleRate = 110000#
.SampleSize = 2048

2. Issue a command to load the signal by doing either of the following:

   – Use the VBT command .LoadSettings

   – Use the LoadSettings microcode in a pattern

For example, the following statement loads the signal MySignal:
.Signals(“MySig”).LoadSettings
IG-XL evaluates the signal to see if it is already loaded.

3. If the signal is not loaded, it is loaded into the signals memory of the UltraPAC80 channel card.

If the signal is already loaded, this step is skipped.

4. The hardware registers of the channel card are updated using a download from signals memory.

The evaluation of whether a signal is already downloaded, the signal download, and the update of hardware registers all add to test time. Obviously, updating hardware for a signal that is already in signals memory is faster than loading a new signal. However, .LoadSettings and the LoadSettings microcode still require that IG-XL evaluate a signal to see if it is new and unlike other signals in signals memory.
The VBT command .ReloadSettings allows you to eliminate the time to evaluate a signal. This command does not write any settings to signal memory, and it does not check the signal software

data structure to see if it has changed. The command simply executes a download from signals memory to the channel card hardware registers.

For more information, see:

•	LoadSettings

•	ReloadSettings

Capture Sample Rate Versus Bandwidth

The UltraPAC Capture instrument has a digital low-pass filter that begins to roll off at 3/8 of the programmed capture sample rate. This is sometimes referred to as “3/4 Nyquist.”

Choose a capture Fs such that:

$$F_s = \frac{8}{3} \times BW$$

(3–1)

Where BW is the bandwidth of interest.

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UltraPAC80 Capture Programming : UltraCapture Pattern Control

UltraCapture Pattern Control

You can control an UltraCapture instrument using pattern microcodes, allowing changes to instrument states and parameters during a program. You can obtain repeatable timing by programming an instrument using pattern microcodes and Signals programming.

You create a pattern independently of the test program using the PatternTool or a text editor. Patterns are composed of a series of numbered digital vectors. These digital vectors define the data for each channel listed in the pattern, as well as the pattern microcode used for controlling the instrument during pattern execution.

The following topics are available:

- UltraCapture Pattern Microcodes
- Understanding UltraPAC Pattern Interlocking
- UltraCapture ASCII Pattern Syntax

☑ **Note:**

To use pattern microcodes to control the UltraCapture on UltraFLEX, you must include a digital pin in the pin map, the channel map, and the pin list of the pattern module.

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UltraPAC80 Capture Programming : UltraCapture Pattern Control : UltraCapture Pattern Microcodes

UltraCapture Pattern Microcodes

A microcode controls a single function, such as starting the capture of a signal. A series of extended microcode commands are available for use inside a pattern. Patterns allow you to control the precise timing of when a signal is captured. To establish repeatable timing, or to synchronize with other instruments (such as the digital subsystem) in the tester, use a pattern. If you want to start capturing a signal but are not concerned with precise timing, however, then you do not need a pattern.

The table lists the extended microcodes used for issuing commands to an UltraCapture instrument from a pattern.

UltraCapture Pattern Microcodes

Microcode	Description
Connect <Connection Set_Name>	Instructs an UltraCapture to switch its connections to the specified connection set. Predefined ConnectionSets are: • ToDUT: On the selected instruments, disconnects any existing connections and connects the instrument(s) to the DUT. • Disconnect: On the selected instruments, disconnects any existing connections. See Connection Sets.

Disconnect	Instructs an UltraCapture to reset all connections to their disconnected state.
Trig <SignalName>	The instrument immediately starts capturing the specified capture signal. This microcode takes an optional argument, which is the name of the signal to be captured. If the signal name is unspecified, the default signal is used. See TheHdw.UltraCapture.Pins.Signals.DefaultSignal.
Resync	Forces phase alignment of an UltraCapture instrument clock with the HSD T0 clock.
Enable_Inst_Cond	Enables the instrument condition bit. The UltraCapture asserts the instrument condition bit to the pattern generator at the completion of a capture.
Disable_Inst_Cond	Disables an UltraCapture from asserting the instrument condition bit.
Enable_Alarm	Enables all UltraCapture alarms and opens the alarm window for UltraCapture in a pattern. This microcode reopens the alarm window that had been previously issued a Disable_Alarm microcode.
Disable_Alarm	Disables all UltraCapture alarms and closes the alarm window for UltraCapture in a pattern.
LoadSettings <SignalName>	Instructs an UltraCapture to load the settings for the specified capture signal. SignalName is optional. The default signal is used if no SignalName is included.

☑ **Note:**

UltraPAC80 microcodes have specific rules regarding the necessary spacing in vectors between microcodes on a specific channel. See UltraPAC80 Specifications.

UltraCapture does not use PSets. However, it provides the equivalent capability using Signals programming in VBT and the LoadSettings microcode in a pattern.

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UltraPAC80 Capture Programming : UltraCapture Pattern Control : Understanding UltraPAC Pattern Interlocking

Understanding UltraPAC Pattern Interlocking

By default, hardware for UltraPAC instruments (UltraCaptures and UltraSources) is set to listen to programming commands that are initiated from VBT and that originate from the host computer. When a digital pattern is burst that contains UltraPAC microcodes, such as LoadSettings and Connect, the hardware is activated to also listen for instructions that originate from the digital pattern generator which are sent through the Instrument Sync Link (ISL) bus.

While an UltraPAC instrument is set to listen for ISL instructions, IG-XL locks out all commands originating from VBT, and a run-time error is generated if you attempt to program the UltraPAC instrument from VBT while the pattern is running. This pattern interlocking is done to avoid collisions in the hardware in the case where instructions from different sources occur too close together for the hardware to react properly to both instructions.

The lockout starts when a digital pattern containing LoadSettings and Connect or Disconnect microcodes is started. The lockout is released at end of a Digital.Patgen.Haltwait VBT statement indicating that the pattern burst is complete or when an explicit Patgen.Halt method is executed.

At times, however, a test program may be written such that VBT commands are to be issued to an UltraPAC instrument while a pattern is running. For example, you might need to issue commands while a pattern is looping on a wait for CPU flag. In such cases, you can override the pattern lockout to allow VBT programming to occur. The syntax is:

TheHdw.UltraCapture.Pins("MyPin").PatternMicroCodeControl = tlOff

Once the needed VBT commands are issued, you should restore .PatternMicroCodeControl to tlOn before continuing pattern execution.

Note that changing the .PatternMicroCodeControl property only toggles the VBT interlock mechanism. It does not disable ISL commands. Collisions are still possible when

.PatternMicroCodeControl is set to tlOff and the digital pattern is still running and executing microcodes. Take care when toggling .PatternMicroCodeControl such that this does not occur. For more information, see PatternMicroCodeControl in the VBT Syntax Reference.

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UltraPAC80 Capture Programming : UltraCapture Pattern Control : UltraCapture ASCII Pattern Syntax

UltraCapture ASCII Pattern Syntax

UltraCapture pattern microcodes are instructions you can place on an individual vector of a pattern. For that vector, a microcode instructs the UltraCapture to perform a particular action, such as triggering a new capture.

You can only use UltraCapture microcodes with pins that the instruments statement has specified as UltraCapture. For example:

```
instruments = {
      DUT_AOUT:UltraCapture;
}
```

For more information, see Instruments Statement in the Pattern Language section.

If a vector contains any UltraCapture related microcode other than nop, that vector is placed in SRM.

The format for specifying an UltraCapture microcode on a vector is:

(pin-list : UltraCapture = microcode)

The following code shows ASCII pattern microcode syntax for UltraSource and UltraCapture instruments. The device analog input (DUT_AIN) and analog output (DUT_AOUT) are connected to a UltraSource and a UltraCapture respectively.

```
instruments = {
      DUT_AOUT:UltraCapture;
      DUT_AIN:UltraSource;
}
vector ( $tset, Digital1, Digital2)
{
 > tset1  0        1        ;
```

```
(DUT_AOUT = Resync)
(DUT_AIN = Resync)
> -     1       1       ;
(DUT_AOUT = Trig )
(DUT_AIN = Start )
> -     0       1       ; /* Trigger/Start the default signal */
(DUT_AOUT = Trig Audio_1kHz)
(DUT_AIN = Start Audio_1kHz)
> -     1       1       ; /* Trigger/Start a named signal */
(DUT_AOUT = Enable_Alarm)
(DUT_AIN = Start1 )
> -     0       1       ;
(DUT_AOUT = Disable_Alarm)
(DUT_AIN = StartE )
> -     1       1       ;
(DUT_AOUT = Enable_Inst_Cond)
(DUT_AIN = Start1E )
> -     0       -       ;
(DUT_AOUT = Disable_Inst_Cond)
(DUT_AIN = Stop)
> -     1       -       ;
(DUT_AIN = StopE)
> -     0       -       ;
(DUT_AOUT = LoadSettings)
(DUT_AIN = LoadSettings)
> -     1       -       ;
> -     -       -       ;
}
```

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UltraPAC80 Capture Programming : Using UltraPAC80 in Concurrent Testing

Using UltraPAC80 in Concurrent Testing

To support concurrent testing, you can assign each UltraPAC80 instrument channel (UltraCapture or UltraSource) to a unique time domain. Concurrent features for UltraPAC80 include:

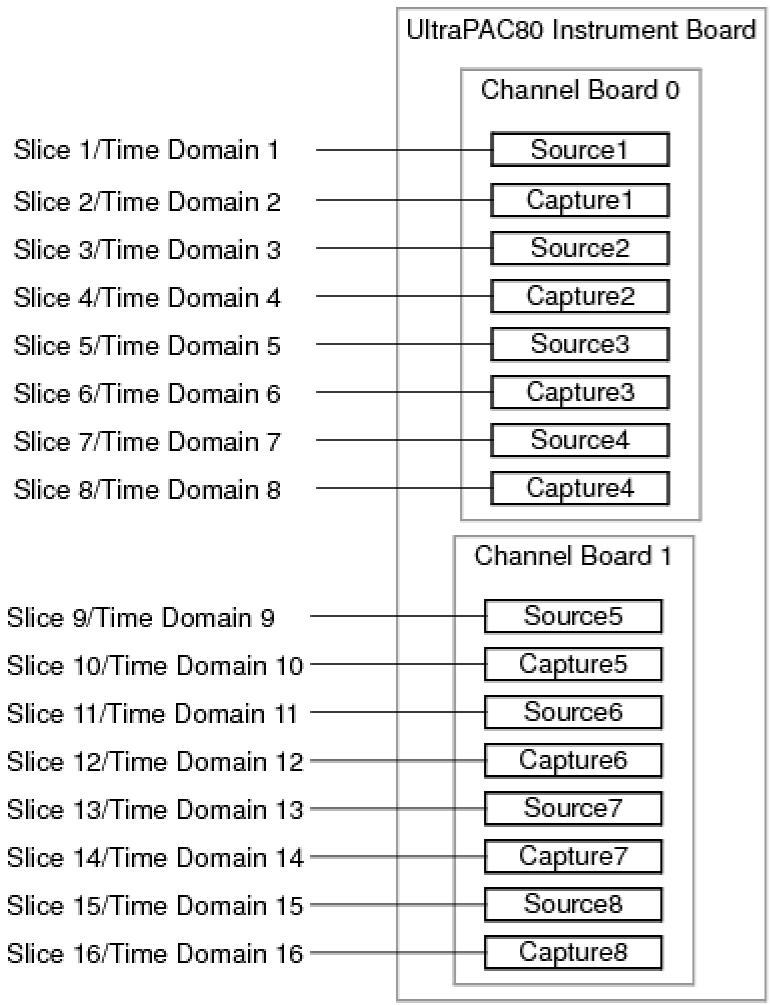
- Up to 16 instruments per channel card can be configured in up to 16 time domains:
  - Eight UltraSource instruments per channel card
  - Eight UltraCapture instruments per channel card
- Each UltraPAC80 instrument can be controlled by unique time domains simultaneously using Sync-Link. Features include:
  - Independent access to Sync-Link
  - Independent sample clock programming
  - Independent clock synchronization with other UltraPAC80 instruments or with digital clocks
  - Independent control by digital pattern
  - Independent capture memory and triggers

- Multiple instrument channels can be grouped into a common flow domain

- UltraPAC80 fully supports flow domains and asynchronous pattern starts

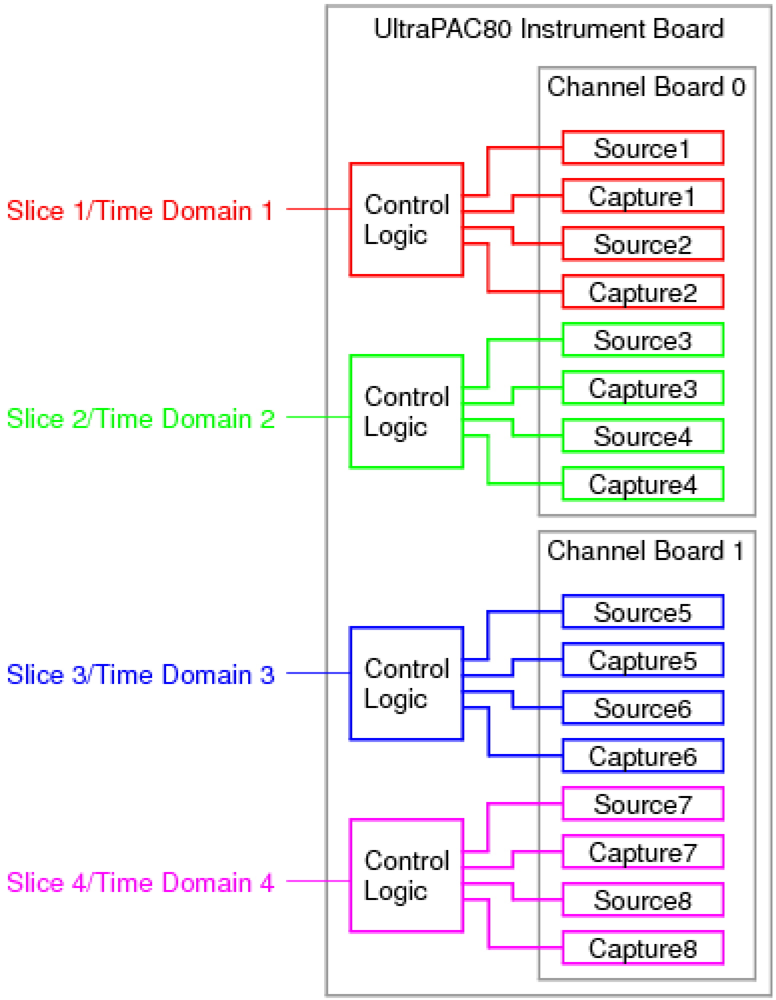
Note that when a shared UltraPAC80 resource is used within multiple concurrently executed flows, only one flow can communicate with the shared resource at a given time. Only the pattern associated with the time domain for which the ApplyLevelsTiming was last applied will be “active” and capable of communicating with the UltraPAC channel through the Instrument Sync Link (ISL), and only if the UltraPAC channel is also assigned to that time domain. When ApplyLevelsTiming is executed and a MOSC change occurs, IG-XL resolves that time domain and sets up appropriate ISL linkages between digital channels/PatGen, MOSC, and analog instrumentation that are all in the same time domain. If a pattern bust associated with a different time domain, which is not currently the “active” time domain, attempts to communicate with an UltraPAC instrument, then a BaseIslacHardware timeout error is returned. If this happens, adjust the sequencing of patterns and ApplyLevelsTiming calls such that the most recently executed ApplyLevelsTiming sets the correct time domain for the subsequent pattern burst.

The following figure shows a configuration in each UltraPAC80 instrument assigned to its own time domain, providing the maximum of 16 time domains.

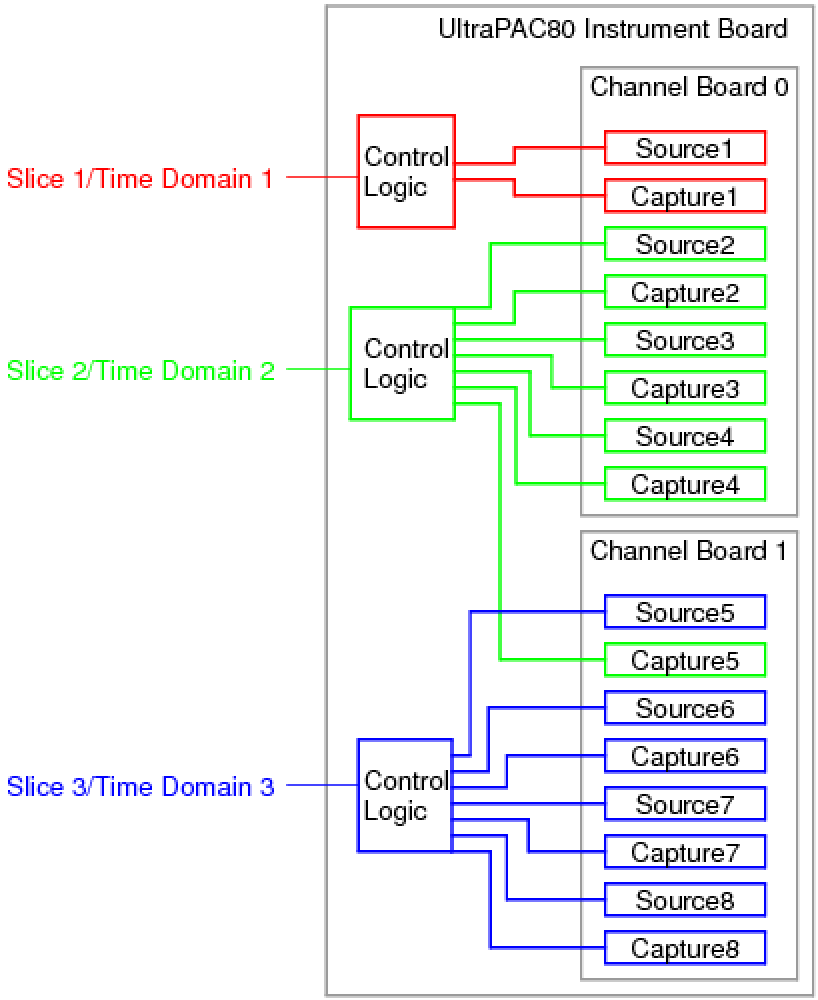


UltraPAC80 Concurrent Testing Configuration Example, 16 Time Domains

However, you can also group several UltraPAC80 instruments into a common time domain. The following figures show examples of UltraPAC80 instruments configured for four and for three time domains.



UltraPAC80 Concurrent Testing Configuration Example, 4 Time Domains



UltraPAC80 Concurrent Testing Configuration Example, 3 Time Domains

For more information, see About Concurrent Test.

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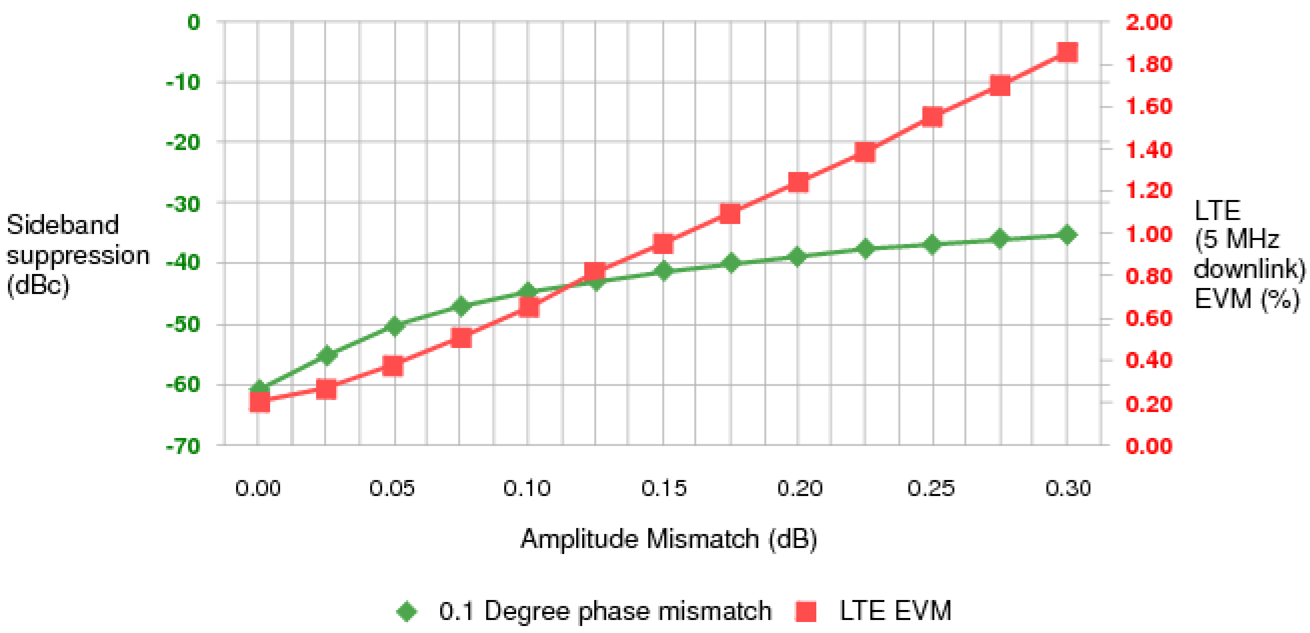
UltraPAC80 Capture Programming : Achieving IQ Alignment for UltraCapture

Achieving IQ Alignment for UltraCapture

Poor sideband suppression degrades modulation metrics in wireless testing, and sideband suppression is completely determined by the alignment of in-phase and quadrature signal components (IQ alignment).

As an example, the following graph shows the relation of sideband suppression and amplitude mismatch to error vector magnitude (EVM) for LTE. As the graph shows, increasing amplitude mismatch rapidly decreases sideband suppression.



Sideband Suppression and LTE EVM for 0.1 Degree Phase Mismatch

The following topics are available:

- IQ Calibration Method
- Capturing IQ Signals

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UltraPAC80 Capture Programming : Achieving IQ Alignment for UltraCapture : IQ Calibration Method

IQ Calibration Method

An UltraPAC80 instrument achieves IQ alignment using two types of calibration:

- Factory calibration stored on the instrument board
- Run-time IQ calibration performed when a signal is loaded

Factory Calibration for IQ Alignment

Factory calibration for IQ alignment is done by Teradyne and is fixed when an UltraPAC80 instrument is produced. The calibration values are stored on the EEPROMs for each instrument board. This calibration provides rough phase alignment (to within a few hundred picoseconds)

If IQ calibration is enabled when you load a signal, IG-XL determines whether that signal has been calibrated. If not, IQ calibration is performed, and factory calibration values (along with real-time values) are applied at that time.

Run-Time IQ Calibration

IG-XL includes a focused calibration routine that you can use on a per signal basis. This IQ calibration provides a final adjustment to the phase specification to account for components that vary over time and temperature, along with amplitude alignment.

You can run IQ calibration and apply it to any signal. If enabled, IQ focused calibration is performed at Signals.Item().LoadSettings or at pattern start, where the pattern contains a LoadSettings microcode for the signal.

By default, IG-XL checks to see if IQ focused calibration was previously executed for the specified signal on the instrument pair since job load. If calibration was executed, IG-XL applies the amplitude and phase correction factors to the capture instrument pair for this signal. Otherwise, IG-

XL starts IQ calibration to phase-match the pair of instruments and amplitude-match the instruments to the signal's voltage range and calibration frequency of interest.

To use run-time IQ calibration, the channels that make up an I/Q pair must be adjacent. You can use the following channel pairs: 1/2, 3/4, 5/6, or 7/8. I/Q channel pairs that do not adhere to this rule return a run-time error when you try load a signal with IQ calibration enabled. See the example in Capturing IQ Signals.

Note the following:

- IQ calibration only applies to the .Bandwidth and .VoltageRange parameters specified during a Signal's .LoadSettings. Modifying these parameters after .LoadSettings is applied results in degraded alignment.
- IQ calibration uses the frequency setting to obtain an optimal alignment for that frequency. Instrument phase is adjusted for optimal alignment.
- No IQ matching is done during DSP preprocessing.
- Running the IQ calibration uses the associated UltraSource instruments. Each UltraSource instrument is restored when the run-time calibration is complete. However, the UltraSource must not be running when .LoadSettings is called on the UltraCapture.

For more information, see UltraCapture in the VBT Syntax Reference.

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UltraPAC80 Capture Programming : Achieving IQ Alignment for UltraCapture : Capturing IQ Signals

Capturing IQ Signals

The following is an example programming procedure for capturing an IQ signal. This procedure:

- Makes two captures, one for the I signal and one for the Q signal, that are phase-matched and amplitude-matched
- Analyzes the captured data in a DSP procedure

The procedure is as follows:

1. From a loaded test program, define two pins for the I and Q signals. For example, the channel map could be:

Example IQ Channel Map

Pin Name	Type	Site 0	Comments
IPin	UltraCapture	<slot>.cappos1	1st capture instrument
QPin	UltraCapture	<slot>.cappos2	2nd capture instrument

2. Define a signal, enabling IQ calibration to phase-match and amplitude-match the two capture instruments to perform the capture:

```
With TheHdw.UltraCapture.Pins("IPin, QPin").Signals.Add("IQCapture")
  ' <Settings of instrument>
```

```
  .VoltageRange = <Signal voltage range> ' Used for amplitude matching
  .SampleRate = <Signal sample rate> ' Program together to align captures
  .Calibration.IQEnabled = True
  .Calibration.FrequencyOfInterest = <Fi of signal to be captured in Hz>
End With
```

3. Load the instrument settings of the Signal:

```
TheHdw.UltraCapture.Pins("IPin, QPin").Signals("IQCapture").LoadSettings
```

Once the settings are loaded, IG-XL determines if IQ focused calibration was previously executed for this signal on the instrument pair since this job load. If so, IG-XL applies the amplitude and phase correction factors to the capture instrument pair for this signal. Otherwise, IG-XL starts IQ calibration to phase-match the pair of instruments and amplitude-match the instruments to the specified calibration frequency of interest and signal's voltage range.

After IQ calibration, all user settings and connections are restored to their previous states.

4. Connect the instruments to the DUT:

```
TheHdw.UltraCapture.Pins("IPin, QPin").Connect
```

5. Perform an IQ capture:

```
TheHdw.UltraCapture.Pins("IPin, QPin").Signals("IQCapture").Trigger(True)
```

IG-XL triggers the two capture instruments synchronously.

6. Associate a DspWave with the capture:

```
Dim wave As New DspWave
wave = TheHdw.UltraCapture.Pins("IPin, QPin").Signals("IQCapture").DspWave
```

7. Create a DSP procedure to process the captures:

```
Public Function ProcessIQCapture(ByVal IWave As DspWave, ByVal QWave As _
DspWave, ByRef dspResult As Double)
  ' Optionally, you can combine I & Q captures to ComplexDouble DspWave
  Dim IQWave As New DspWave
  IQWave = IWave.ComplexCombine(tldspRealWithImaginary, QWave)
```

```
  dspResult = ...
End Function
```

8.	Execute the DSP procedure:

```
Dim dspResult As New SiteDouble
rundsp.ProcessIQCapture(wave.Pins("IPin"), wave.Pins("QPin), dspResult)
```

For more information, see UltraCapture in the VBT Syntax Reference.

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UltraPAC80 Capture Programming : Programming the Relay Counter

Programming the Relay Counter

The mechanical relays used in the UltraCapture have a finite life span and can perform a limited number of switches before they begin to fail. To maximize the life span of the instrument, you may want to minimize the number of times a relay is switched during a test program run. IG-XL includes a RelayCounter feature for the UltraCapture that allows you to collect relay switching data.

All relay switches are counted whether initiated from the tester computer using VBT or from the pattern with a microcode.

Relay counting is unaffected by resets (those before or after a test program or resets you initiate), but counts do not persist beyond the current program. Loading a new test program or restarting DataTool resets relay switching counts. Counts are enabled by default.

For example, you have a test program loaded that includes two UltraCapture pins (Cap1 and Cap2) with two sites each. You want to determine how many times each UltraCapture relay switched during a program run. You can do the following:

1. Open an immediate window and executes code to begin relay counting:

```
TheHdw.UltraCapture.Pins("Cap1, Cap2").RelayCounter.Disabled = False
TheHdw.UltraCapture.Pins("Cap1, Cap2").RelayCounter.Start
```

IG-XL initializes or resets its user relay counting variables.

Note that this is a convenient time to start relay counting, but generally no sites are active or selected at this time. This method (and all RelayCounter features) ignores site status and selection.

2. Run the test program or loop it several times.

IG-XL counts all switches of mechanical relays from all captures (calibration, reset, pattern, and tester computer).

3. When the test program completes its runs, execute code in the immediate window to generate a relay counting report and save it to a text file:

```
TheHdw.UltraCapture.Pins("Cap1, Cap2").RelayCounter.DumpCurrentCount
"UltraCapture_Relay.txt"
```

Each count includes the number of times a relay has switched since the Start call in step 1, including auto and run-time calibrations.

IG-XL then generates a formatted text file that includes all relays associated with the UltraCapture (in rows) and data for each pin/site specified in the pins selection (in columns) regardless of site activity/selection. The report is formatted in a fixed-width font and tab delimited to allow you to paste it into an Excel worksheet.

Here is an example of a relay report showing four UltraCapture instruments:

****************** Dump Relay Counter for UltraCapture *********************

Pin	cap1	cap1	cap2	cap2
Site	0	1	0	1
Slot.Chan#	2.CapPos1	2.CapPos2	2.CapPos3	2.CapPos4
CONNECT_HR_POS	0	0	0	0
CONNECT_HR_NEG	0	0	0	0
CAL_POS_DUT	0	0	0	0
CAL_NEG_DUT	0	0	0	0
CONNECT_PPMU_POS_DUT	0	0	0	0
CONNECT_PPMU_NEG_DUT	0	0	0	0
CONNECT_POS_DUT	0	0	0	0
CONNECT_NEG_DUT	0	0	0	0
ATTEN_HR1_9DB	0	0	0	0
ATTEN_HR1_6DB	0	0	0	0
ATTEN_HR2_3DB	0	0	0	0
ATTEN_HR1_0DB	0	0	0	0
GAIN_HR_3DB	0	0	0	0
GAIN_HR_9DB	0	0	0	0
GAIN_HR_0DB	0	0	0	0
ATTEN_HR2_0DB	0	0	0	0
ATTEN_HF_BIT0	0	0	0	0
ATTEN_HF_BIT1	0	0	0	0
ATTEN_HF_BIT2	0	0	0	0
ATTEN_HF_BIT0_GND	0	0	0	0

ATTEN_HF_BIT1_GND	0	0	0	0
ATTEN_HF_BIT2_GND	0	0	0	0
BASELINE2_GND	0	0	0	0
BASELINE2_DGS	0	0	0	0
CONNECT_HR_NEG_DGS	0	0	0	0
CONNECT_HR_POS_DGS	0	0	0	0
BASELINE1_GND	0	0	0	0
BASELINE1_DGS	0	0	0	0
CONNECT_HF_POS	0	0	0	0
CONNECT_HF_NEG	0	0	0	0
TERM_HF_NEG	0	0	0	0
TERM_HF_POS	0	0	0	0
GAIN_HR_6DB	0	0	0	0
LPF_HF_30MHz	0	0	0	0
LPF_HF_75MHz	0	0	0	0
GAIN_HF2_9DB	0	0	0	0
GAIN_HF2_0DB	0	0	0	0
GAIN_HF1_6DB	0	0	0	0
GAIN_HF1_3DB	0	0	0	0
GAIN_HF1_0DB	0	0	0	0

*************** End of Dump Relay Counter for UltraCapture *****************

You can also insert calls to count relay switches and to generate reports into a test program.

For more information, see Pins.RelayCounter in the VBT Syntax Reference.

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UltraPAC80 Capture Theory of Operation

UltraPAC80 Capture Theory of Operation

This section provides conceptual and reference information about the UltraPAC80 Capture instrument's hardware and signal path. The following topics are available:

For information on UltraPAC80 Capture instrument performance specifications and standards, see UltraPAC80 Specifications.

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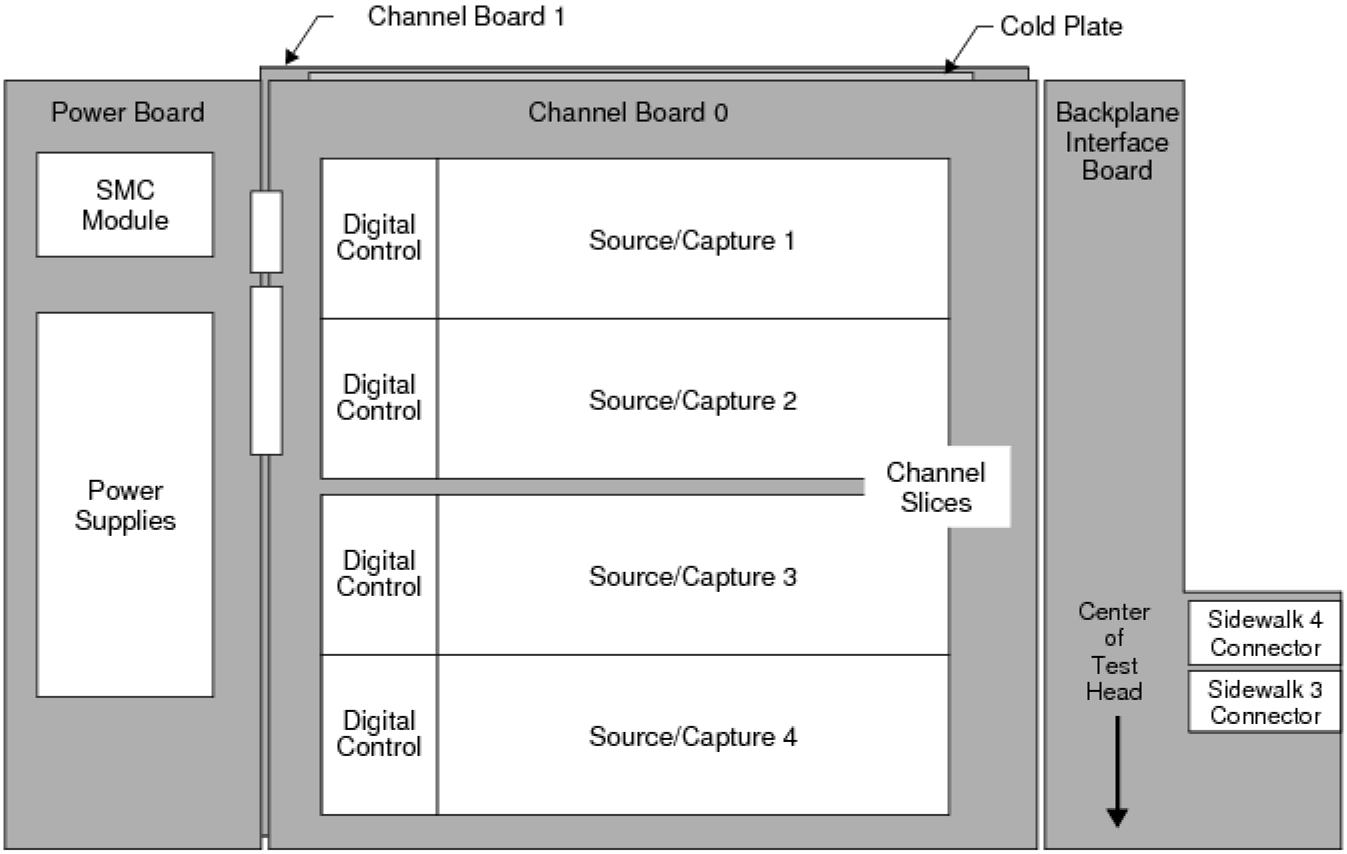
UltraPAC80 Capture Theory of Operation : UltraPAC80 Instrument Board

UltraPAC80 Instrument Board

The UltraPAC80 instrument board assembly occupies a single slot in the UltraFLEX test head and contains eight UltraPAC80 Source and eight UltraPAC80 Capture instruments. The assembly consists of:

- Two identical channel boards mounted on opposite sides of a liquid-cooled cold plate.
  - Each channel board has four UltraSource channels and four UltraCapture channels, for a total of eight source and eight capture per instrument board
  - Each UltraSource and UltraCapture channel consists of: Pos pin, Neg pin, a PPMU resource, DGS, and a converter (ADC for UltraCapture, DAC for UltraSource)
  - Each UltraSource and UltraCapture channel has a high frequency (HF) and a high resolution (HR) path
- A backplane interface board (BIB) that provides connections to the UltraFLEX backplane and manages signal flow between the test system and instrument boards
- A power board that converts the +48V system power to the various on-board power voltages that the UltraPAC80 board requires

The following figure shows the layout of an UltraPAC80 instrument board.



UltraPAC80 Instrument Board Hardware Block Diagram

UltraPAC Channel Boards

The UltraPAC channel boards contain the primary UltraPAC80 instrument control and tester communication components as well as the analog input and output stage of the UltraPAC80 instruments. Each channel board also has the following features:

- An interface to communicate with the databus of the tester computer
- 32 Pin Parametric Measurement Unit (PPMU) instruments, which can be used for parametric testing. Each UltraSource and UltraCapture instrument has two PPMU pins, one for each positive and each negative pin.
- MoveLink connections from the UltraPAC80 Capture instruments to one or more dedicated DSP computers in the test system for high speed transfer and DSP processing of acquired capture data
- Instrument Sync-Link (ISL) connections from each UltraPAC80 channel to the Pattern Generator in the High Speed Digital instrument for accurate and repeatable synchronization

- A differential internal analog loopback connection shared between a pair of UltraPAC Source and a pair of UltraPAC Capture instruments
- Four device ground sense (DGS) connections shared by the source, capture, and PPMU circuits for a pair of source and a pair of capture instruments

UltraSource and UltraCapture Channels

Each UltraPAC80 instrument board has eight UltraSource channels and eight UltraCapture channels in total. An UltraSource channel can drive AWG signals with bandwidths from DC to 80MHz to device analog input pins. An UltraCapture channel can capture signals from device output pins with bandwidths from DC to 75MHz.

Eight digital control FPGAs, four on each channel board, support the source and capture hardware. Each FPGA supports an UltraSource/UltraCapture channel pair.

Each UltraSource and UltraCapture channel also includes the following resources:

- Precision 24-bit high resolution AWG source and capture
- 16-bit high speed AWG source and capture
- Gain ranging
- Filtering
- Digital sample rate conversion
- Source and Capture and Signals memory
- Connection to PPMU per input/output pin

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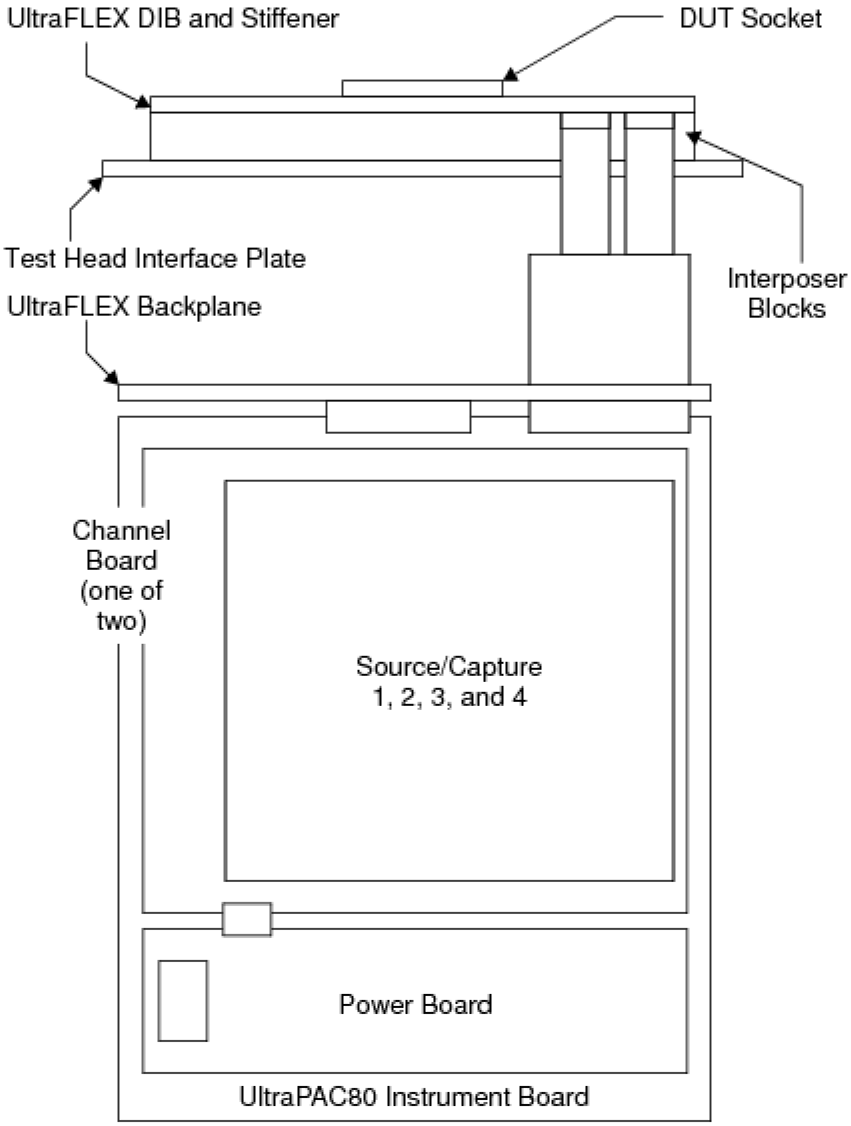
UltraPAC80 Capture Theory of Operation : UltraPAC80 Instrument Board : Test Head Connections

Test Head Connections

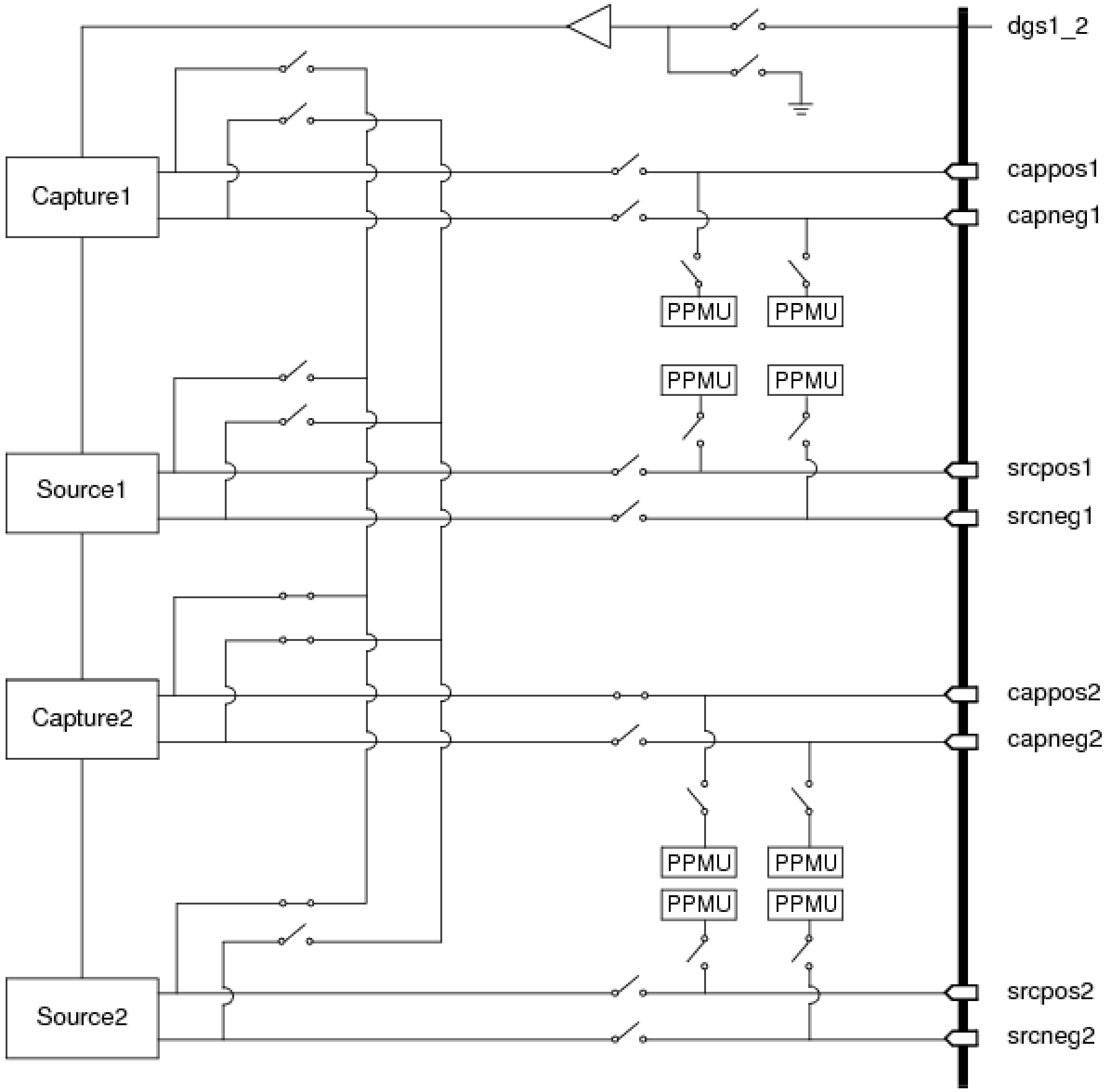
This section describes the connections of UltraPAC boards to the UltraFLEX test head backplane. The following figure shows how the instrument board is connected through interposer blocks to deliver power to the DUT.



UltraPAC80 Instrument Board Connection to the DUT

Up to eight UltraPAC80 boards can be installed in an UltraFLEX test head. An UltraPAC80 can be installed in any available slot in the test head. Channels are connected to the DIB using two interposer blocks occupying sidewalks 3 and 4 of the loadboard end.

The following figure shows a connection model for the UltraPAC80 signals to the DUT.



UltraPAC80 Channel Connections to the DUT

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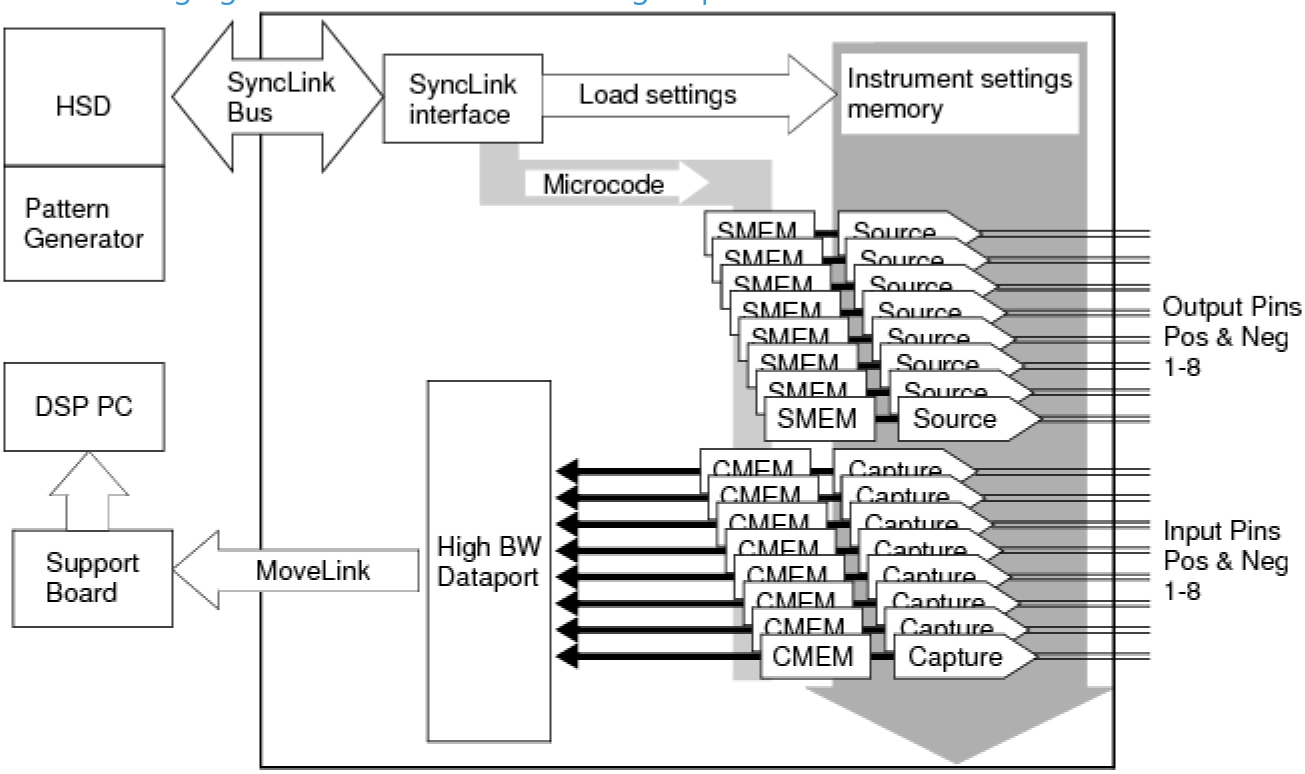
UltraPAC80 Capture Theory of Operation : UltraPAC80 Capture Signal Path

UltraPAC80 Capture Signal Path

This section describes the architecture of the UltraCapture channel. Each of the eight Capture channels on the UltraPAC80 instrument board is identical.

The following figure shows the UltraPAC80 signal paths at the board level.

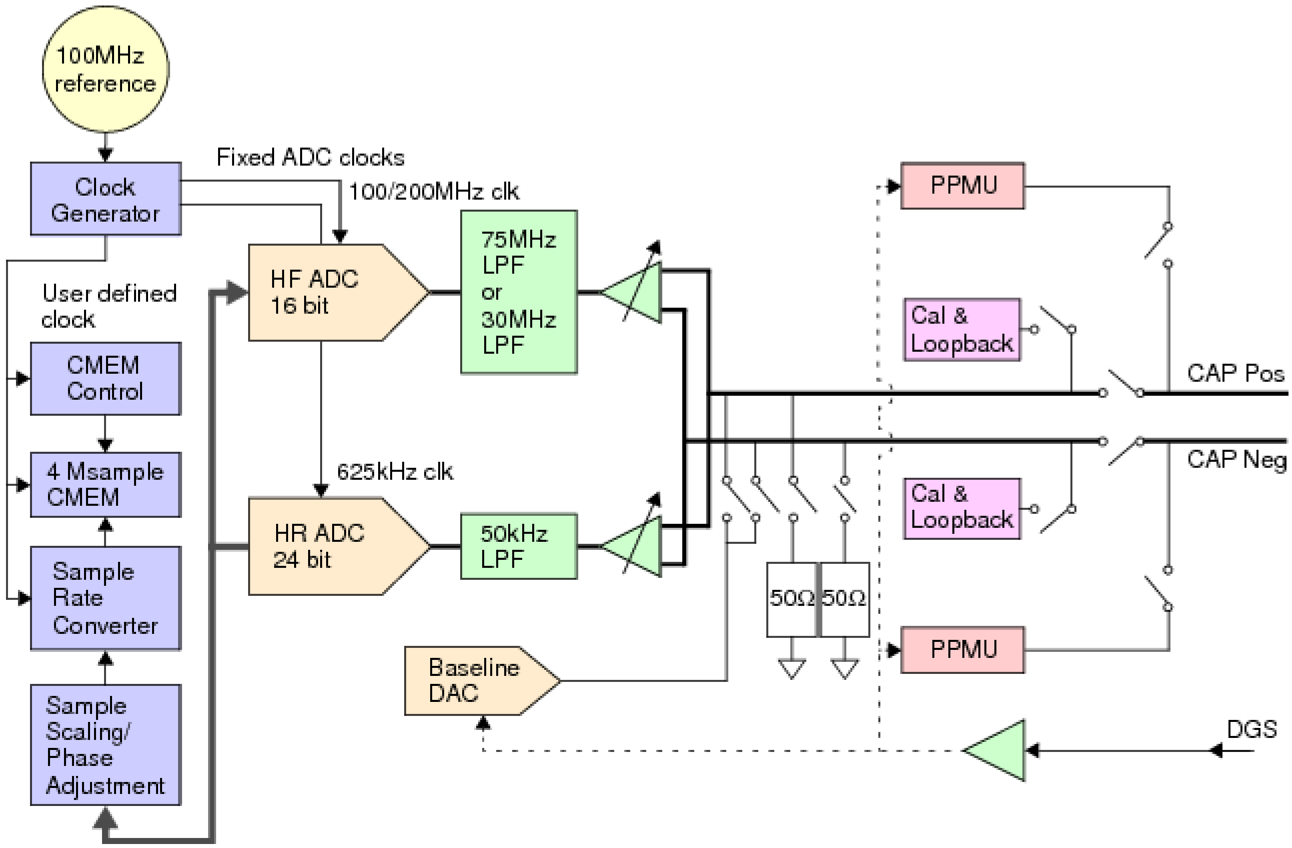


UltraPAC80 Board Diagram

The UltraPAC80 board includes eight independent UltraCaptures, each with separate high resolution (HR) and high frequency (HF) signal paths. The HR path covers the frequency range of DC - 50kHz, while the HF path operates from DC - 75MHz. You can switch input termination on the HF path between 50 ohms and high impedance (HiZ) mode.

The use of two separate signal paths allows UltraCaptures to address a diverse set of tests over a wide range of speed and performance requirements. The high resolution (HR) path uses 24-bit based ADCs and is used for low frequency capture applications such as audio, static (ramp) linearity measurements, and precision lower speed analog-to-digital converters. The high frequency (HF) path uses 16-bit ADCs and is used for general purpose capture for high speed ADC, video ADCs, wireless baseband, and baseband converters.

The next figure shows a block diagram of a single UltraPAC80 Capture channel. The following sections describe individual components of the block diagram.



UltraPAC80 Capture Channel Block Diagram (one of eight shown)

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UltraPAC80 Capture Theory of Operation : UltraPAC80 Capture Signal Path : Clock Generator

Clock Generator

The UltraPAC contains a clock generator circuit (one circuit common to all channels), which generates fixed clocks from the test system 100MHz master reference clock. These clocks are used to drive both the HF and HR source ADCs. The clocks are not user programmable. User defined sample rates are achieved using sample rate conversion. (See Digital Signal Processing.)

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UltraPAC80 Capture Theory of Operation : UltraPAC80 Capture Signal Path : Signal Delivery and Input State

Signal Delivery and Input State

An input signal enters an UltraPAC80 Capture instrument as an analog voltage signal from the DUT. The signal is an arbitrary waveform that must have an amplitude less than the programmed voltage range and frequency components less than the instrument's bandwidth to be accurately captured. Signal delivery also includes DGS input pins, which are used to provide the device's ground voltage reference to the instrument card. All voltages measured with UltraPAC are always the programmed level relative to the DGS voltage (in the case of single-ended measurements). DGS lines should typically be grounded close to the DUT, but they can be grounded near the interposer for low current applications. A single DGS input is shared amongst two source and two capture channels: 1-2, 3-4, 5-6, 7-8. DGS pin to channel routing is fixed and is not user selectable.

Each UltraPAC Capture positive, negative, and DGS line incorporates a corresponding shield. The shield is used to isolate the source signals from EMF interference as they pass through coaxial cabling from the DIB to the instrument boards via the interposer interface.

The shields also carry return currents for the capture input lines. Therefore, make sure to ground the shields on the DIB. The best practice for the grounding these shields is application dependent. Typically, they are grounded to the analog DIB ground plane near the interposer connections.

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UltraPAC80 Capture Theory of Operation : UltraPAC80 Capture Signal Path : Termination

Termination

The input of an UltraPAC Capture has an optional 50 ohm termination, which you can select. If it is not selected, the input of an UltraCapture is high impedance. This optional termination is available only when using the High Frequency path. The High Resolution path is always a high impedance termination.

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UltraPAC80 Capture Theory of Operation : UltraPAC80 Capture Signal Path : Offset

Offset

The offset circuitry of an UltraPAC80 Capture allows you to maximize the dynamic range of the instrument by removing dc offset from the capture signal before amplification and gain ranging stages. You can remove common mode offset from a differential signal, and you can also remove dc offset from a single-ended signal. You cannot remove differential offset from differential signals.

When measuring signals with dc offset removal, an UltraPAC Capture automatically reapplies the offset to the captured waveform samples when moving samples to the DSP. In this way, waveform samples are an exact representation of the actual voltages present at the device output. The dc baseline offset removal voltage is always relative to DGS.

For more information on UltraPAC80 Capture compliance, see UltraPAC80 Specifications.

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UltraPAC80 Capture Theory of Operation : UltraPAC80 Capture Signal Path : Voltage Ranging

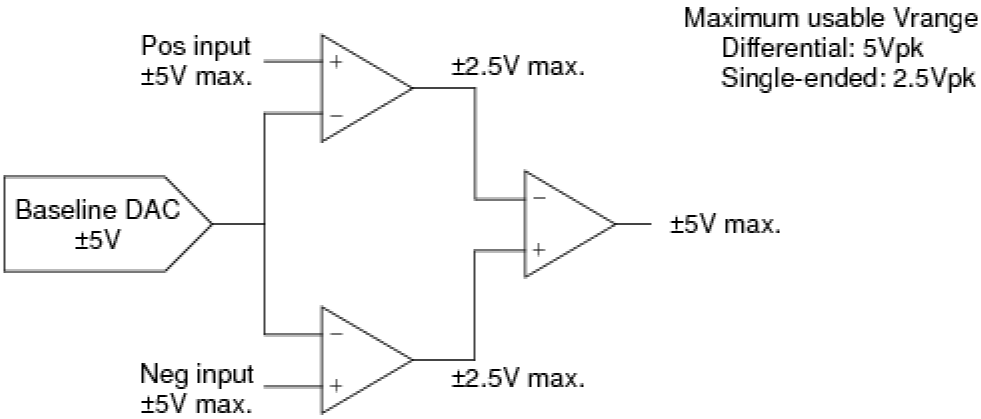
Voltage Ranging
Voltage range specifies the maximum or minimum voltage that can be applied between the positive and negative pins, or differential voltage. For example, if the voltage between the positive and negative pins is 1.28V, the range must be set to 1.28V or higher to accommodate the signal. Setting a capture to a range smaller than the maximum voltage causes the capture to clip the signal, resulting in an overrange alarm and clipped waveform samples.
Another constraint to consider is that the voltage on either the positive or negative pins must be less than the instrument compliance (~5V for HR mode, ~4 V for HF mode, with respect to DGS).
UltraPAC80 Capture ranges are as follows:

- High Resolution mode: 0.442Vpk-dff, 0.625Vpk-dff, 0.884Vpk-dff, 1.25Vpk-dff, 1.768Vpk-dff, 2.5Vpk-dff, 3.536Vpk-dff, and 5.0Vpk-dff

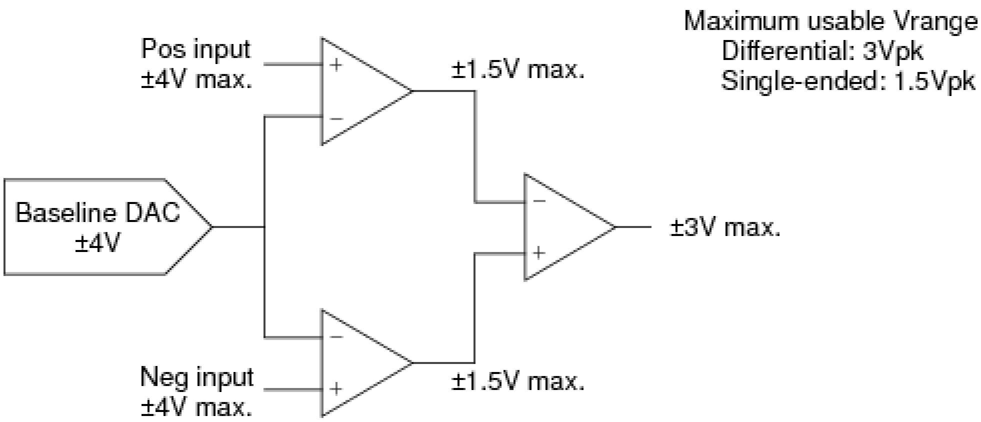
- High Frequency mode: 0.265Vpk-dff, 0.375Vpk-dff, 0.53Vpk-dff, 0.75Vpk-dff, 1.061Vpk-dff, 1.5Vpk-dff, 2.121Vpk-dff, and 3.0Vpk-dff

The UltraPAC80 input stage has several compliance voltages at various amplifiers in the input stage. Input voltages must be within the compliance voltages to avoid clipping one of the analog front end amplifiers.
In single-ended operation, the compliance of the input buffer, not the voltage range, ultimately limits the maximum amplitudes that may be measured. The figure below describes the compliances for each of the amplifiers in the UltraCapture front end.
Note that when measuring differentially, the individual buffers have the necessary compliance to use the 5V range in HR mode and the 3V range in HF mode. When measuring single-ended signals,

however, the compliance limits the maximum peak amplitude that can be measured to 2.5V in HR mode and 1.5V in HF mode.



Input Compliance, HR Mode



Input Compliance, HF Mode

For more information, see UltraPAC80 Specifications.

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UltraPAC80 Capture Theory of Operation : UltraPAC80 Capture Signal Path : Analog Anti-Aliasing Low-Pass Filter

Analog Anti-Aliasing Low-Pass Filter

The scaled input signal passes through a fixed low-pass filter to eliminate any high frequency components that may cause unwanted aliased components in the captured waveform samples after sampled by the analog-to-digital converters.

The low-pass filter cutoff frequencies are 50kHz for the HR path and either 30MHz or 75MHz for HF path (user selectable). Specifications vary when using either the 30MHz or 75MHz filter for the HF capture path.

By specification, the filters are flat to the cutoff frequency. The cutoff frequency does not refer to the -3dB point.

Since the ADCs always operate at a fixed clock rate, only one preset low-pass filter is required per path, with the passband ending at one-half of the converter’s clock rate.

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UltraPAC80 Capture Theory of Operation : UltraPAC80 Capture Signal Path : Capture ADCs

Capture ADCs

The filtered signal is passed to the analog-to-digital converters (ADCs), which sample the analog waveform at a fixed sample rate.

An UltraPAC Capture uses different ADC architectures for the HF and HR paths:

- The HF path uses 16-bit ADCs. The key feature of this ADC is its high sample rate and bandwidth.

- The HR path uses 24-bit ADCs. The key feature of this ADC is its good noise, SFDR, and THD performance, especially over the audio band. The HR ADC is also very linear for capturing precise voltage ramps for static integral nonlinearity tests.

After conversion, the digital samples are passed to a DSP processing stage.

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UltraPAC80 Capture Theory of Operation : UltraPAC80 Capture Signal Path : Digital Signal Processing

Digital Signal Processing

An UltraPAC80 Capture uses a series of digital signal processing stages to prepare sampled data from the ADCs before moving data to DSP in the test system for analysis by the test program. First, a digital amplitude scaling stage converts the ADC codes to volts by multiplying the samples by the voltage range that was used. Amplitude calibration factors are also applied at this stage.

Next, an optionally programmable time-based skew is applied to the captured waveform samples from -2500ps to 2500ps. You can use this skew to apply custom generated calibrations, for example to remove skew due to DIB components.

Finally, the data is resampled to convert the instrument's fixed sample rate to the user-specified sample rate using a sample rate conversion algorithm. This process occurs in real time as samples are read and moved from the ADCs to capture memory.

A benefit of sample rate conversion is that the sample rate that you define is mathematically determined and, therefore, provides a fine degree of control. Also, the ADCs use a fixed sample frequency that is often much higher than the sample frequency you request. Therefore, the instrument operates in an oversampled mode and quantization noise is spread over a wider bandwidth and eventually removed by filters in the analog path. The minimum programmable sample rate is 5 ksps, and the maximum sample rate is 200Msps (HF mode, 75MHz bandwidth) or 100Msps (HF mode, 30MHz bandwidth) or 266 2/3ksps (HR mode).

The sample rate converter also includes a digital low-pass filter to remove unwanted high frequency components from the raw ADC samples that would otherwise alias into the Nyquist band according to the user defined sample rate. The filter's cutoff frequency is always equal to 3/8 times the user defined sample rate. This is equivalent to 3/4 of the Nyquist frequency. Hence, this is often referred to as the "3/4 Nyquist digital filter." To accurately measure signals, ensure that the maximum frequency components that you wish to measure are less than the 3/4 Nyquist filter cutoff point and program the sample rate of the capture sufficiently high enough to achieve this. By specification, the digital filter is flat to the cutoff frequency. The cutoff frequency does not refer to the -3dB point.

For more information, see:

- Calculating Source Timing
- Mixed Signal Workshop

Clock Resolution

The UltraPAC80 has a period-based clock. Resolution is defined in terms of period resolution and therefore is frequency dependent. UltraPAC80 frequency is governed by the equation:

$$\text{UltraPAC80 Clock Frequency} = \frac{100\text{ MHz} \times 2^{40}}{2^{40} \times (I + J)}$$

(2–1)

Where: $I >= 1$, $0 < J < 2^{40}$

The 40 bits of clock resolution allows both UltraPAC80 Source and UltraPAC80 Capture instruments to achieve any desired frequency to a degree of precision on a par with the digital subsystem. While the UltraPAC80 is not always guaranteed to be coherent with the digital subsystem in the strict sense (a perfect integer ratio relationship), any source or measurement error introduced due to coherence error is typically less than can be measured by the UltraPAC80 analog performance. You can use Mixed Signal Workshop to calculate strictly coherent clocking solutions while obeying sample rate and sample size rules for UltraPAC instruments.

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UltraPAC80 Capture Theory of Operation : UltraPAC80 Capture Signal Path : Capture Memory

Capture Memory

Digital values from the 3/4 Nyquist digital filter are routed to a 4Msample (4194304 samples) capture memory. The UltraPAC80 Capture uses an instrument initiated move (IIM) to automatically move the captured data from the instrument's capture memory (CMEM) to the memory in either a system DSP computer or the host computer for signal analysis. Samples begin moving from capture memory to the DSP computer as soon as they are acquired. The entire captured sample set does not need to finish before moving data to the DSP computer.

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UltraPAC80 Capture Theory of Operation : Connectivity Options

Connectivity Options

The following topics describe the connectivity options for connecting an UltraPAC80 Capture instrument to various resources through the DIB:

- Differential Device Output Connections
- Single-Ended Device Output Connections
- Mixing Differential and Single-Ended Output Connections

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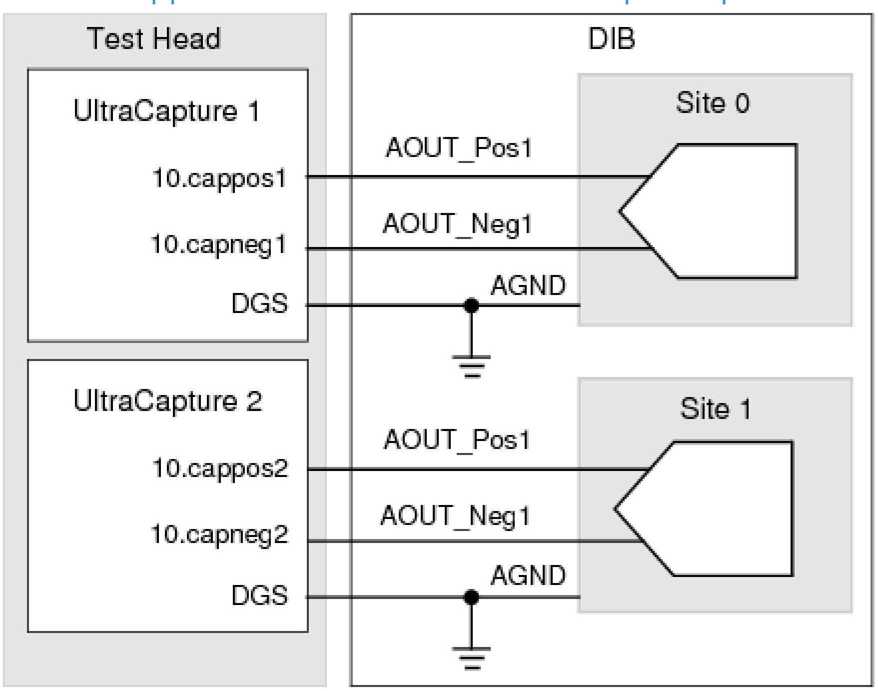
UltraPAC80 Capture Theory of Operation : Connectivity Options : Differential Device Output Connections

Differential Device Output Connections

The following figure shows differential connections for an UltraPAC80 Capture instrument in slot 10. The figure shows two of the eight available UltraCapture channels for this instrument connected to two sites. The capture reference pin in this configuration is connected to the DIB ground. As such, it does not appear as a reference in either the pin map or the channel map, as shown in the tables.



UltraCapture Differential Output Connections

Pin Map for UltraCapture Differential Connections

Group Name	Pin Name	Type
	AOUT_Pos1	Analog

	AOUT_Neg1	Analog

Channel Map for UltraCapture Differential Connections

Pin Name	Package Pin	Type	Site 0	Site 1
AOUT_Pos1		UltraCapture	10.cappos1	10.cappos2
AOUT_Neg1		UltraCapture	10.capneg1	10.capneg2

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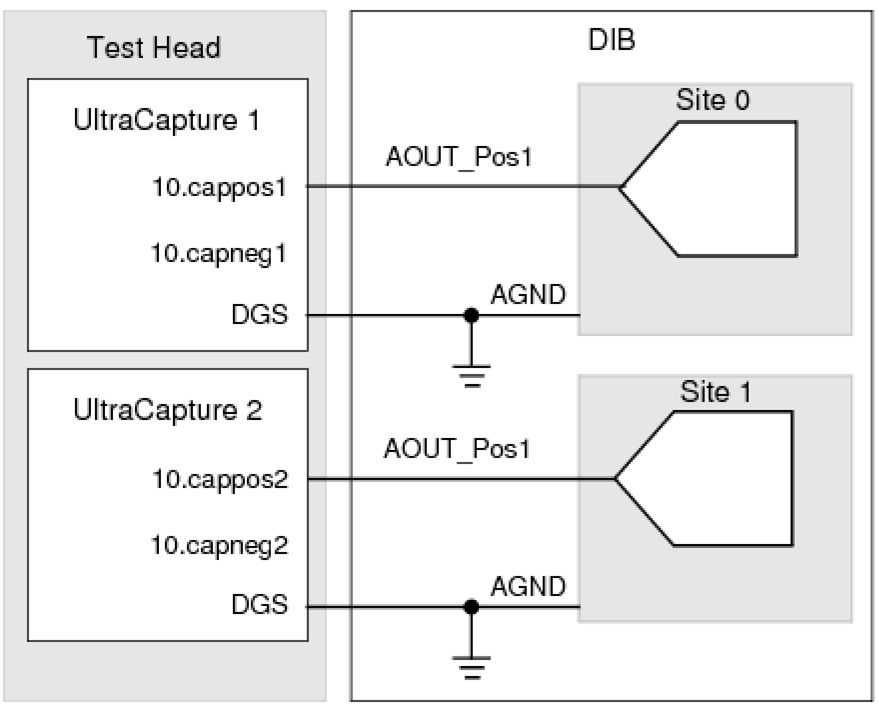
UltraPAC80 Capture Theory of Operation : Connectivity Options : Single-Ended Device Output Connections

Single-Ended Device Output Connections

For connection to single-ended device outputs, DIB circuitry is not required for single-ended to differential conversion. To connect single-ended devices to an UltraPAC80 Capture instrument, connect either the positive or negative pin of the UltraPAC80 differential pin pair to the single-ended device output. The specifications for the positive or negative pins are identical. Note, however, that if you connect to the negative pin, then the measured voltage and waveform samples are inverted.



UltraCapture Single-Ended Output Configuration

The figure shows an UltraCapture instrument in slot 10 with the positive pins connected for two of the eight channels. The negative pins are unconnected.

When measuring a single-ended DUT pin, IG-XL automatically references the measurement to DGS if only one capture pin of the differential pair is connected. You do not need to tie the unused input of the differential pair to DIB ground for this to occur. You can leave it open during the single-ended measurement. If the DUT provides a more suitable voltage reference for the measurement than ground (DGS), you can connect the unused capture pin to this voltage.

You can wire two individual single-ended DUT output pins to each of the UltraPAC80 capture positive and negative pins. Note, however, that you can measure only one device output at a time in this configuration (pin-serial). You can also wire different sites to each of the positive and negative pins of a common UltraPAC80 Capture instrument for single-ended measurements, but you can measure only one site at a time (site-serial).

Pin Map for UltraCapture Single-Ended Connection

Group Name	Pin Name	Type
	A_OUT	Analog

Channel Map for UltraCapture Single-Ended Connection

Pin Name	Package Pin	Type	Site 0	Site 1
AOUT		UltraCapture	10.cappos1	10.cappos2

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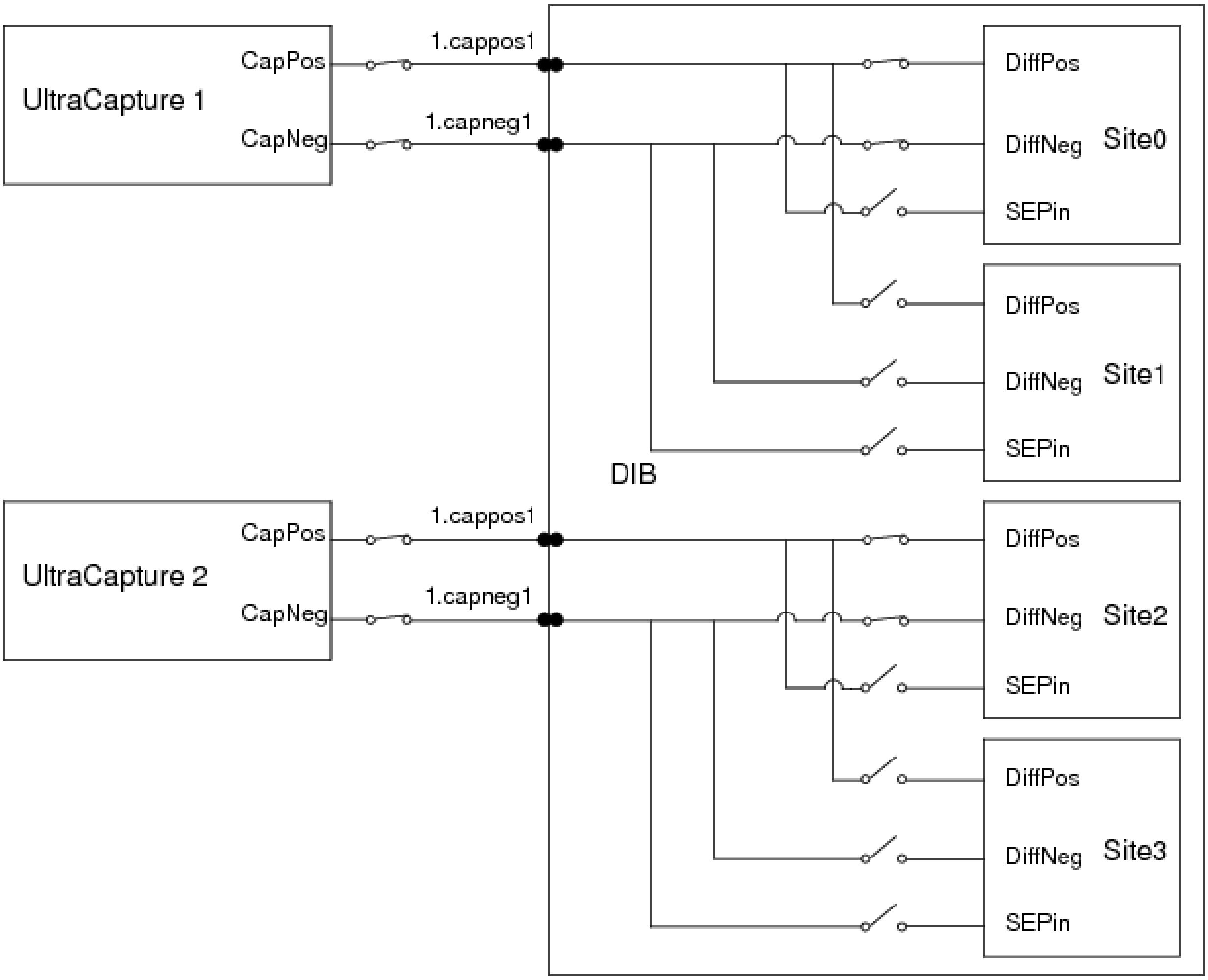
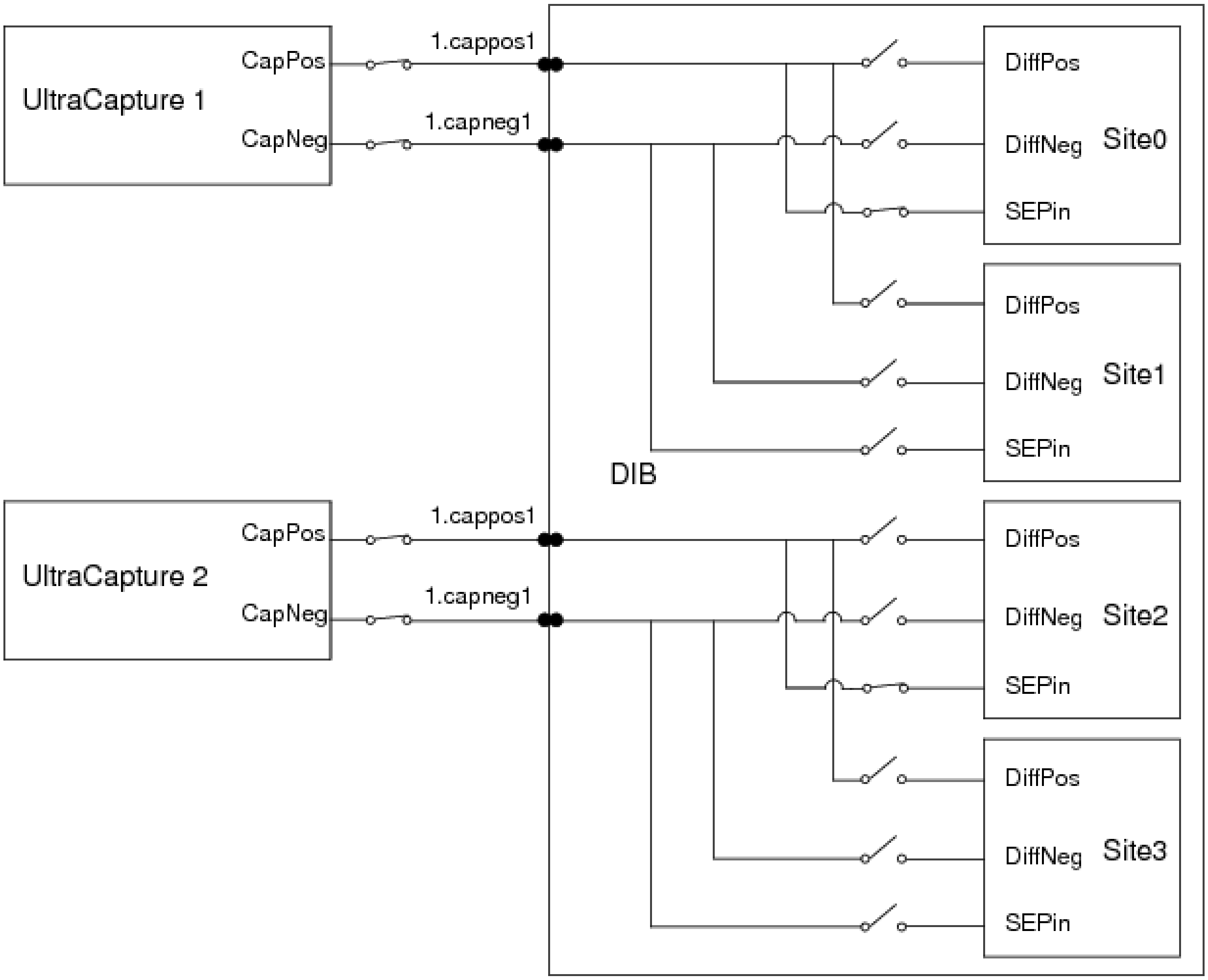
UltraPAC80 Capture Theory of Operation : Connectivity Options : Mixing Differential and Single-Ended Output Connections

Mixing Differential and Single-Ended Output Connections

The following figures show a quadsite program using two UltraCapture instruments in slot 10. Differential pins DiffPos and DiffNeg are shared across sites. UltraCapture 1 is specified behind sites 0 and 1, and UltraCapture 2 is specified behind sites 2 and 3. The same channels are shared by a single-ended pin, SEPin, using each channel (both Pos and Neg) of the UltraCapture instruments. To share the two UltraCaptures among four sites, you need to enter a site loop on even sites and then odd sites.



Quadsite Differential Pins in Even Sites Connected to Two UltraCapture Instruments



Quadsite Single-Ended Pins Connected to Two UltraCapture Instruments

The following is the channel map associated with the example quad-site program.

Channel Map for Quadsite Differential and Single-Ended Pins Sharing Two UltraCaptures

Pin Name	Package Pin	Type	Site 0	Site 1	Site 2	Site 3
DiffPos		UltraCapture	10.cappos1	site0	10.cappos2	site2
DiffNeg		UltraCapture	10.capneg1	site0	10.capneg2	site2
SEPin		UltraCapture	10.cappos1	10.capneg1	10.cappos2	10.capneg2

The VBT syntax to connect the differential pins would be:

TheHdw.UltraCapture.Pins(“DiffPos,DiffNeg”).Connect

The VBT syntax to connect the single-ended pins would be:

TheHdw.UltraCapture.Pins(“SEPin”).Connect

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UltraPAC80 Capture Theory of Operation : Advanced Features

Advanced Features

Signals Memory

UltraPAC80 instruments use a hardware-based state engine to store instrument settings and connections in memory. This is similar to the PSet memory for other UltraFLEX instruments.

The UltraPAC80 does not use the PSet programming model. Instead, signals that are defined in software are loaded and stored to hardware on execution of the .LoadSettings signal method. First, the state of the instrument settings is resolved and instrument settings for that state are stored to specialized signals memory in the UltraPAC hardware. Next, IG-XL sends an instruction to execute that setup. At any later time when a .LoadSettings is executed for this signal, if the signal has not changed, a command is issued to have the hardware execute the state change from signals memory instead of setting each hardware parameter one at a time. This feature provides fast instrument setup changes from VBT and from pattern microcode (LoadSettings pattern microcode).

Additionally, you can use the VBT language .ReloadSettings to apply previously defined signals setups without first resolving the software signal setup. The benefit of the .ReloadSettings command is that it executes even faster than .LoadSettings. However, you cannot use this command to load newly defined or altered signals definitions from software to hardware.

For more information, see:

•	LoadSettings
•	ReloadSettings
•	UltraCapture Pattern Microcodes

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UltraPAC80 Capture Theory of Operation : Advanced Features : Connections Sets

Connections Sets

Connection sets (CSets) are connection state setups that can be defined using VBT programming and stored in specialized CSet memory in the UltraPAC hardware. Once defined and loaded to hardware memory, a state can be applied with a single command either from VBT or under pattern microcode control, thereby allowing for fast instrument connection changes.

An additional benefit of using CSets is that the UltraPAC need not be disconnected at the end of every test to ensure test code modularity. Using the legacy incremental programming model such as .Connect VBT language (which is supported), you must always know the current connection state to apply the appropriate sequence of connects and disconnects to get the instrument to the new desired state. This typically leads test programmers to always put the instrument in a known reset state (disconnected) at the end of every test to be able to effectively reorder test flow.

CSets present a state programming model, not an incremental programming model, hence you only need to set the correct CSet state at the beginning of a test and not worry about reset states to ensure modularity. When a CSet is applied, the UltraPAC hardware automatically resolves the connect state and throws appropriate relays on the board to get to that state. If the board is already in the appropriate state when a CSet is applied, then the hardware does nothing, thereby extending relay lifetime.

For more information, see Connection Sets.

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UltraPAC80 Capture Theory of Operation : User View of DIB Pad Patterns

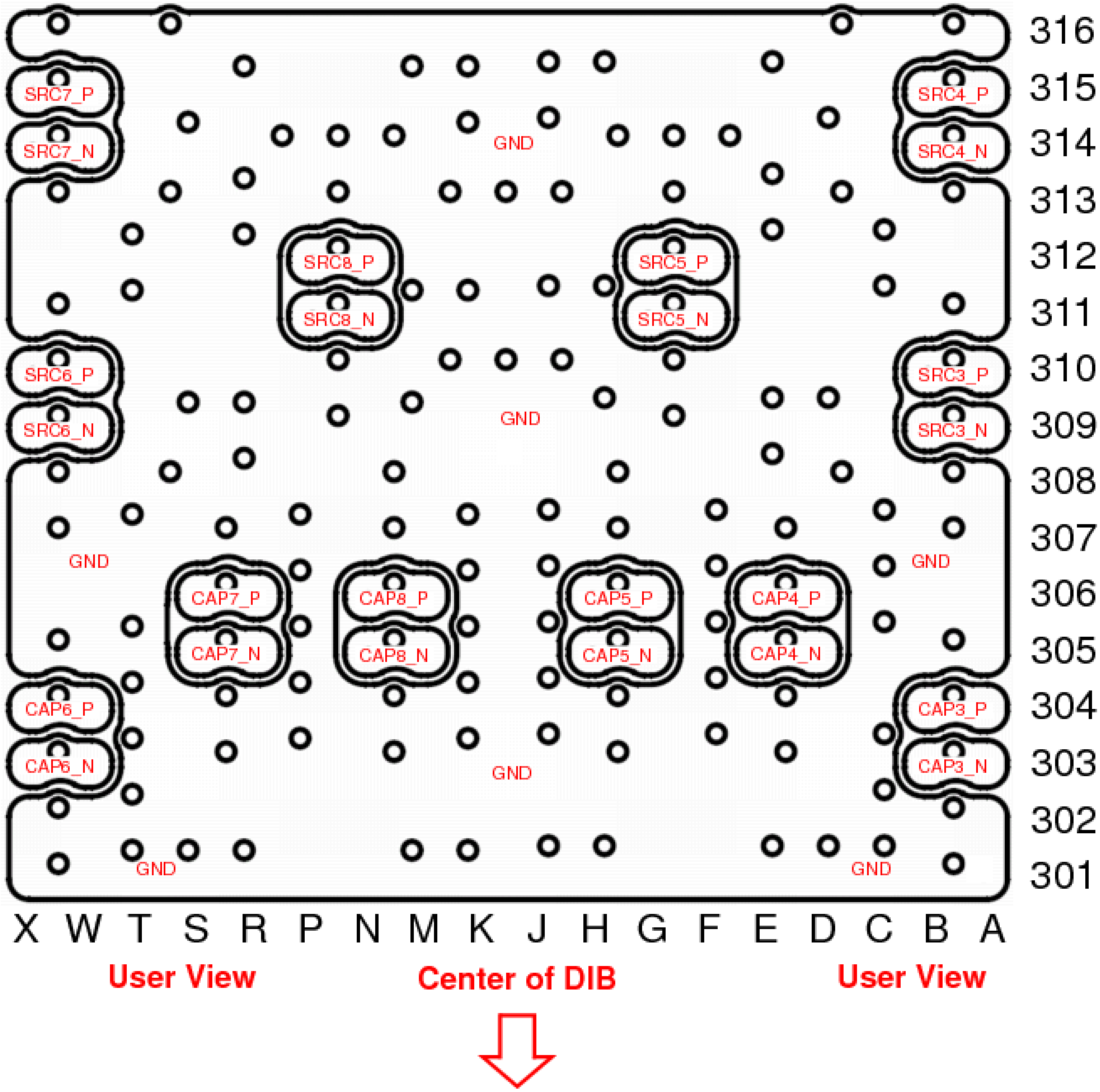
User View of DIB Pad Patterns

The following figures show the arrangement of UltraPAC80 board signals as seen from the user perspective, looking at the topside (device) of the DIB/HIB/PIB and towards the test system. The UltraPAC pinout will work with BBAC and TurboAC pad patterns. However, not all TurboAC and BBAC features are supported on UltraPAC80.

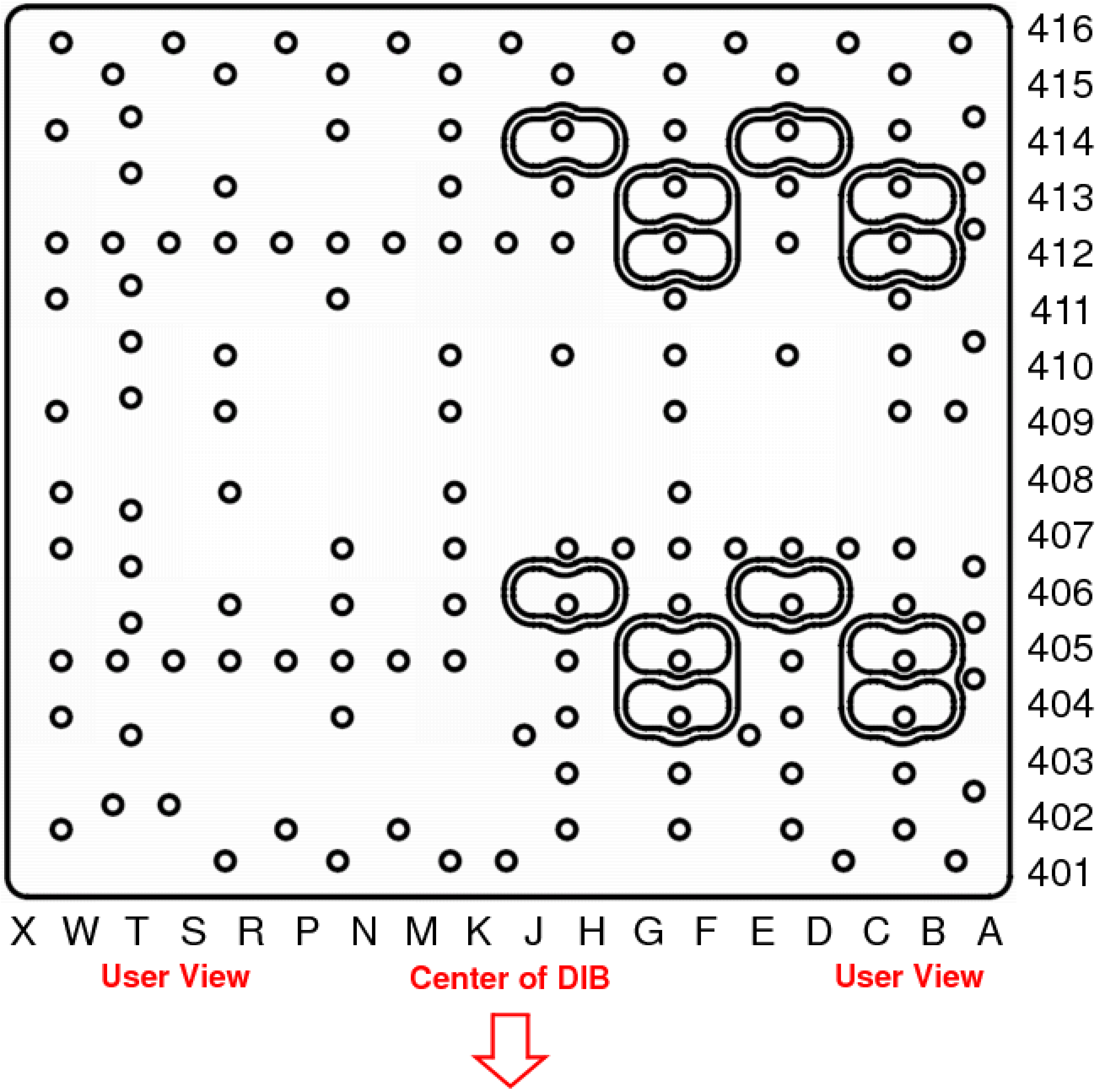
☑ **Note:**

When designing a DIB, use the tester view of the pad patterns (looking at the stiffener side of the DIB/HIB/PIB) because all the artwork and design kits use that view. The UltraPAC80 pad pattern is shown in the UltraPAC80 Signal Pad Pattern (Tester View) section of the UltraFLEX DIB Design Guidelines.

The following figures show the UltraPAC80 user view of the pad patterns for sidewalks 3 and 4 respectively.



UltraPAC80 Pad Pattern for Sidewalk 3 (User View)



UltraPAC80 Pad Pattern for Sidewalk 4 (User View)

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